Payments to grow your world

OPENCHAIN
& Friends

Surviving the “AI Slopaggedon”

How AI shakes the foundations of Open Source

Raphaël Semeteys
2026 March 26

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Raphaël “raphiki” Semeteys

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Head of Tech Advocacy
Senior Architect
Open Source Expert

open source

QSOS project
Projets Libres Podcast
Popularizer in Universities

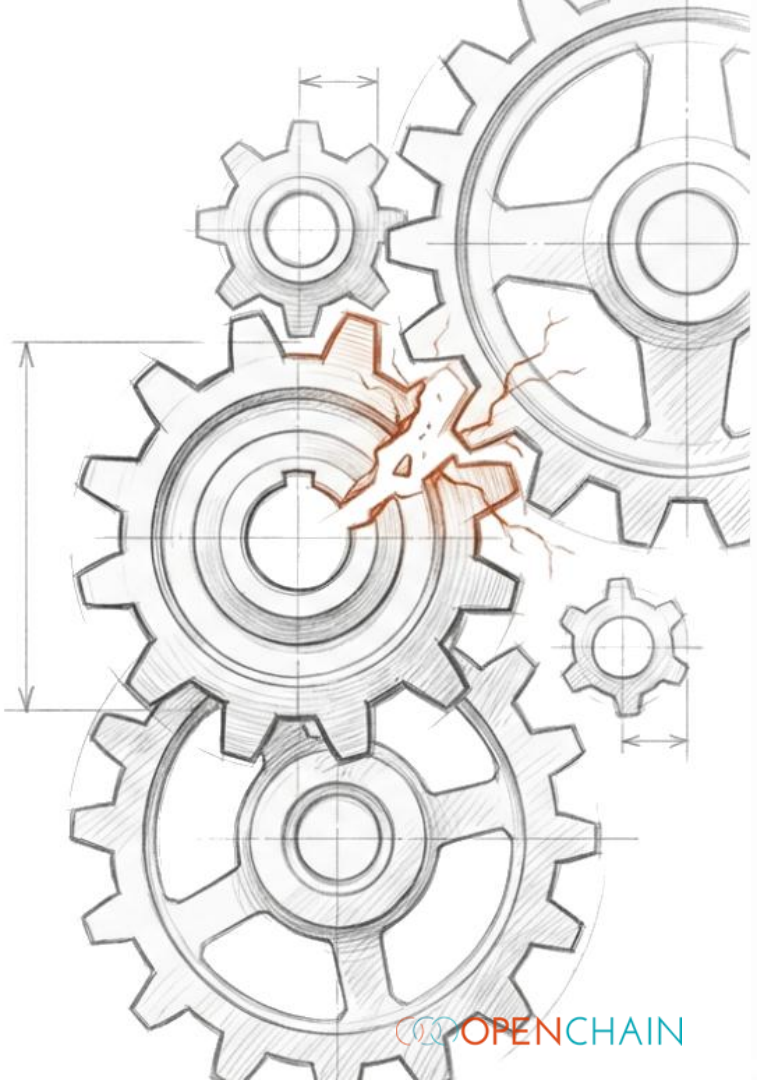
Also

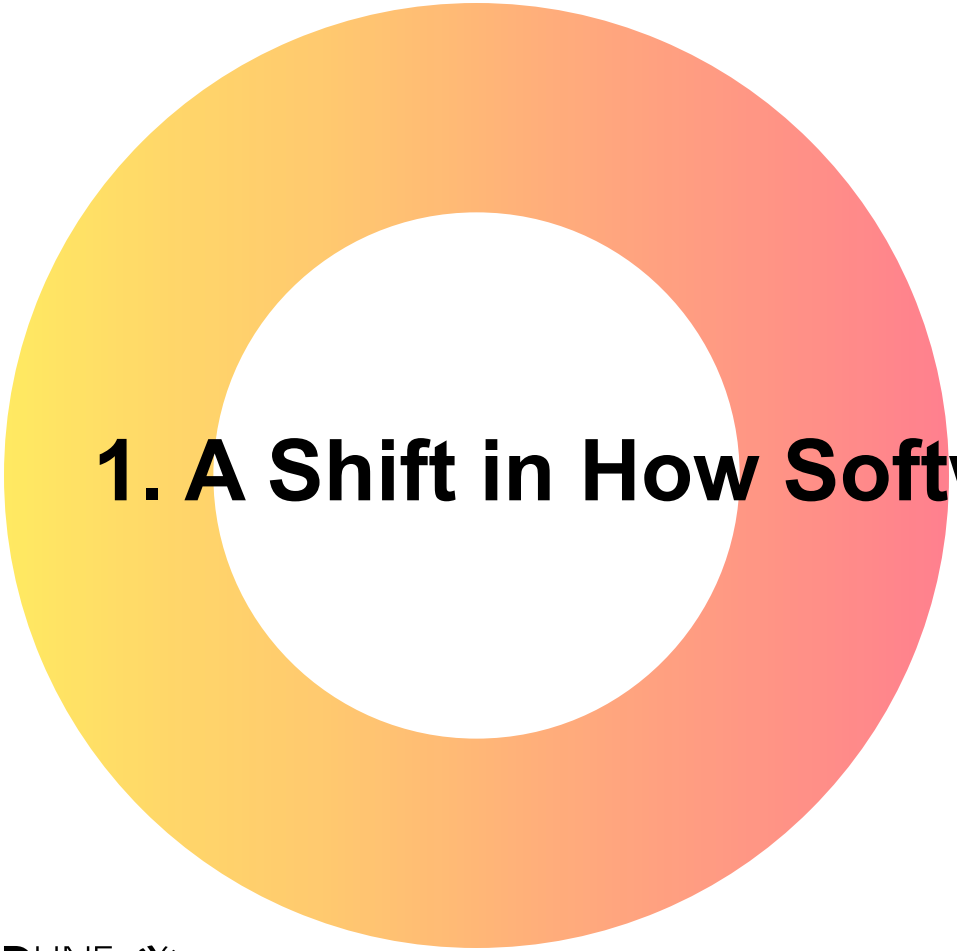
Certified Yoga Teacher - Game Designer
(soon) AI-assisted Novelist



Agenda

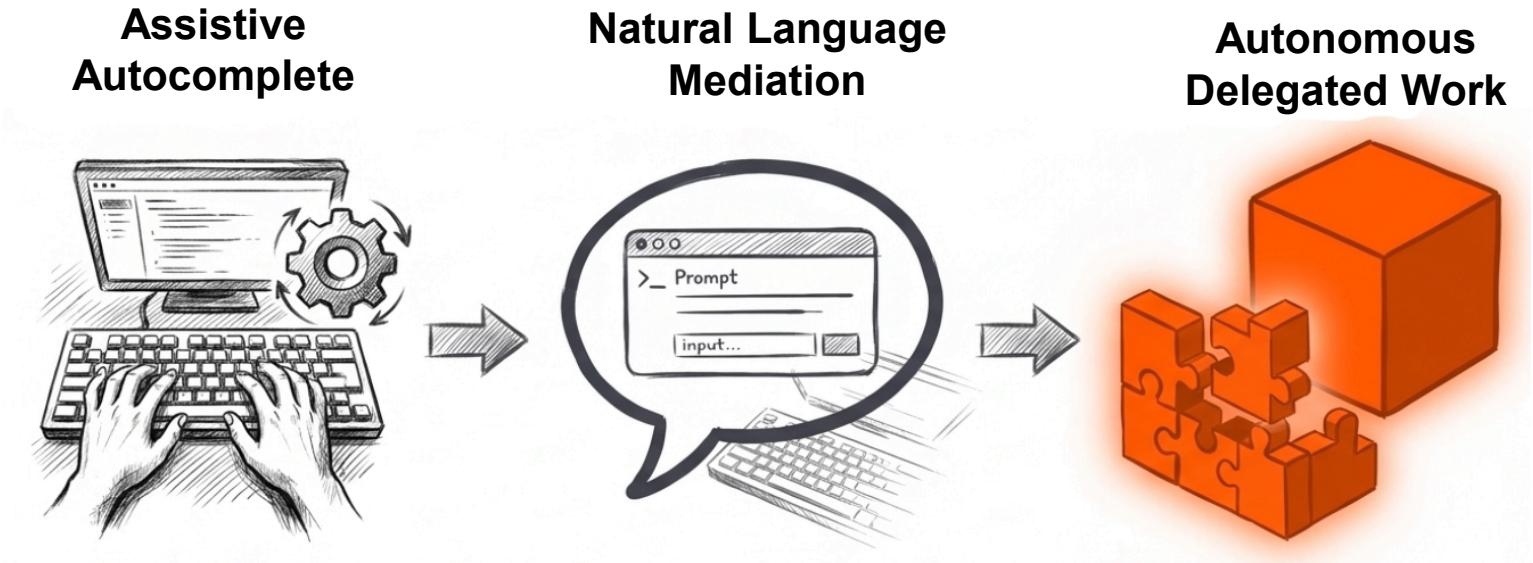
1. A Shift in How Software is Built
2. The First Symptoms we observe
3. Is the System Starting to Break?
4. Adapting to an AI-Native World
5. Open Questions





1. A Shift in How Software is Built

From Code to Prompts to Agents



AI agents assemble open source software mediated entirely by natural language prompts

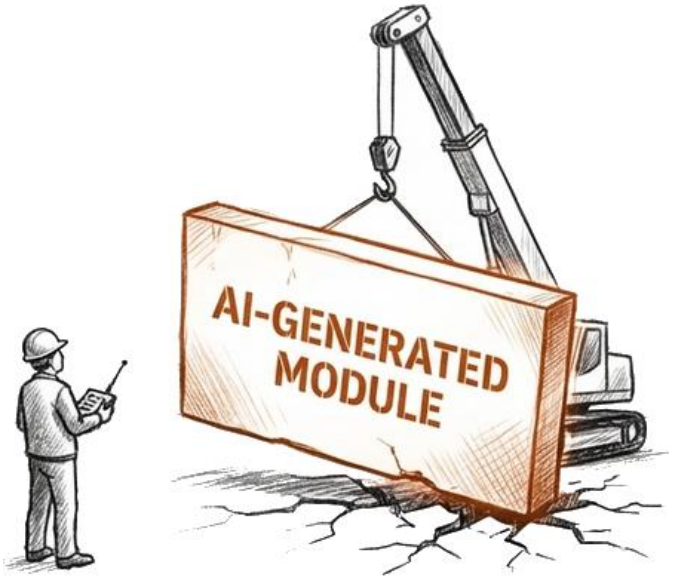
Open Source… without the Community?

Brick-by-Brick Collaboration



Humans submit small patches
And build a community

Concrete Wall Drop

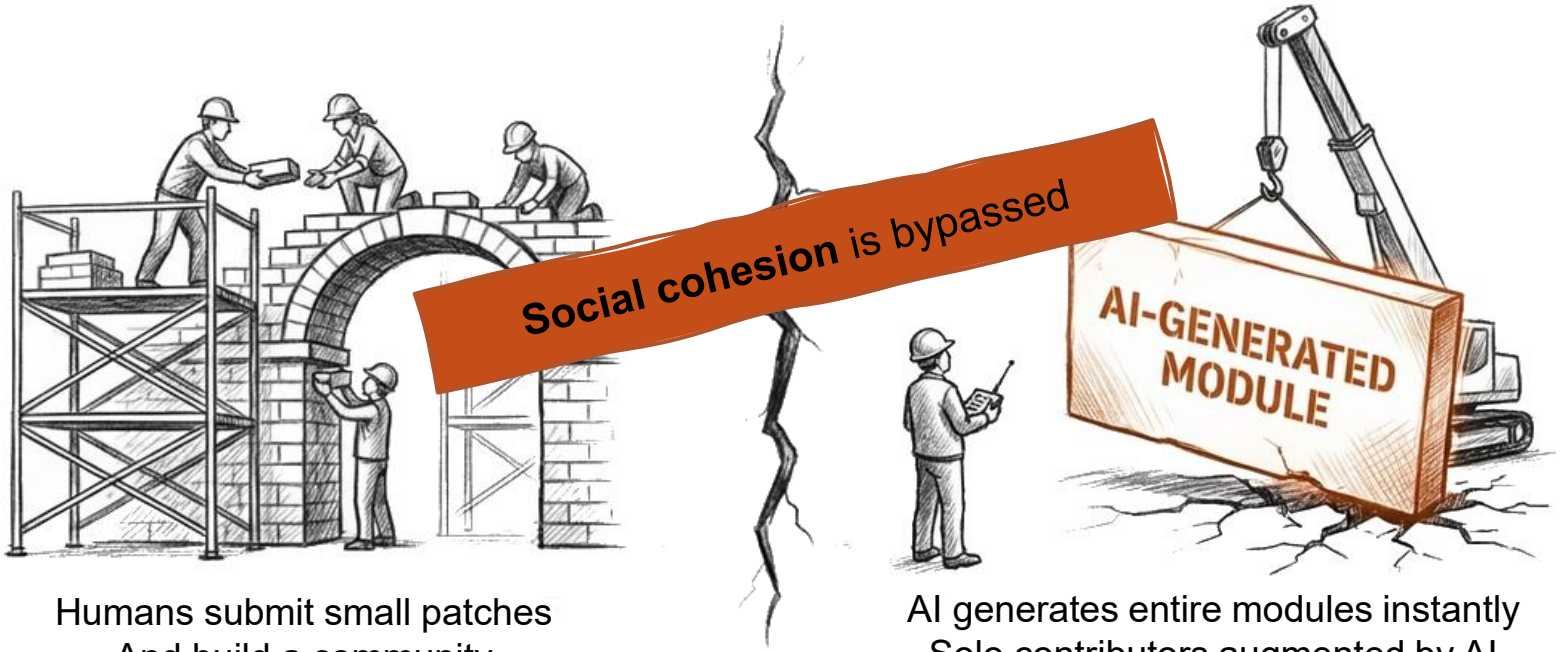


AI generates entire modules instantly
Solo contributors augmented by AI

Open Source… without the Community?

Brick-by-Brick Collaboration

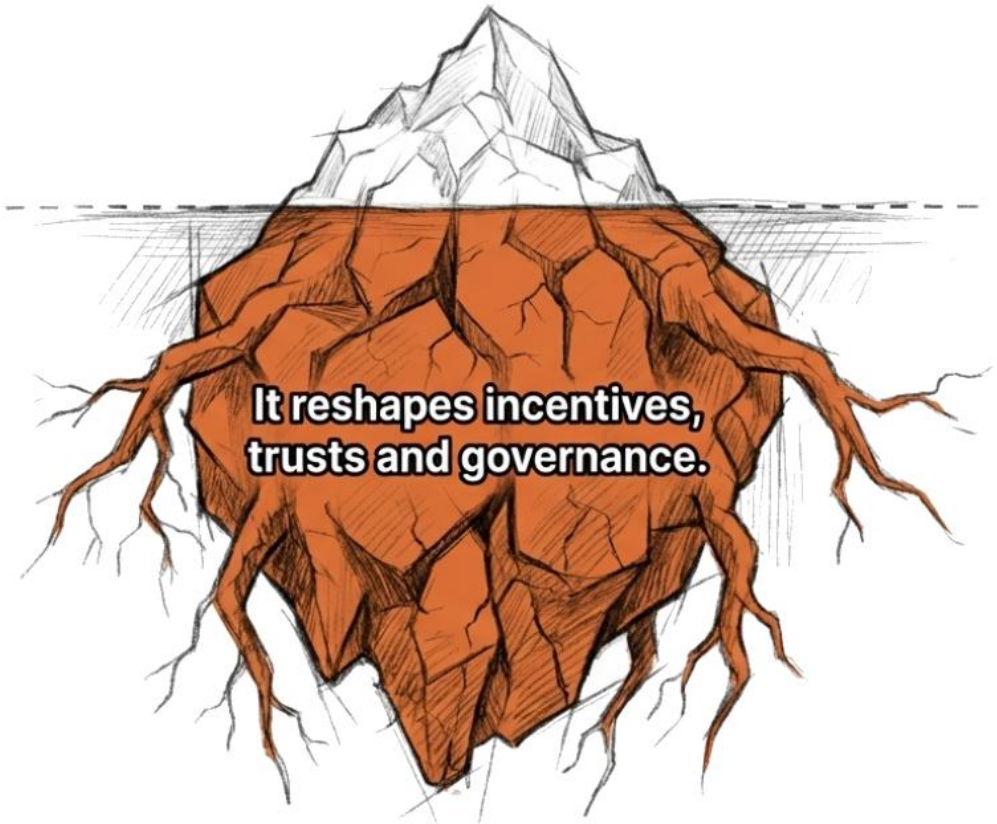
Concrete Wall Drop



Humans submit small patches
And build a community

AI generates entire modules instantly
Solo contributors augmented by AI

This isn’t just a tooling shift





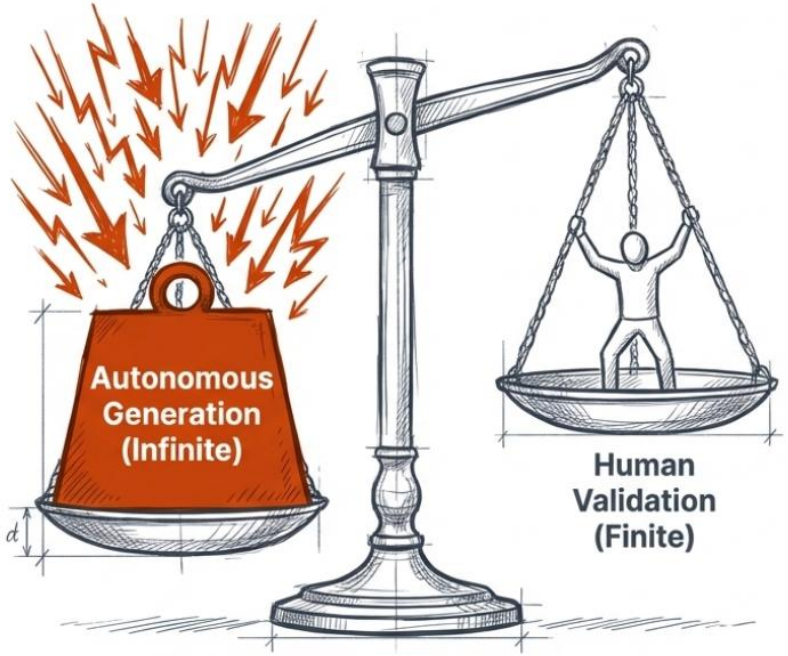
2. The First Symptoms we observe

Code is infinite. Review is not.

Unleashed Contributions

No friction to generate code

AI can instantly generate entire modules



Validation Bottleneck

Cognitive heavy lifting still fall entirely on humans

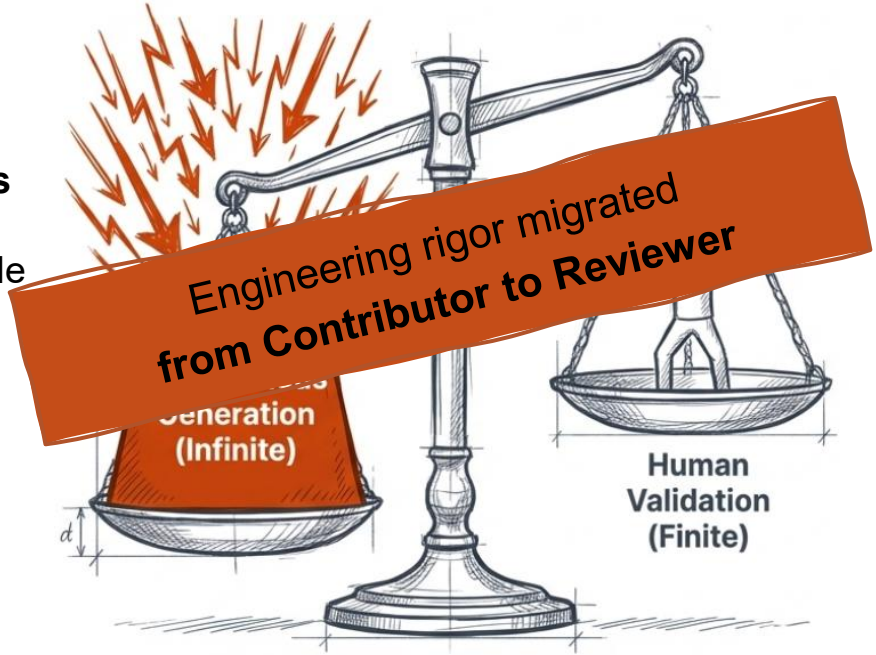
Continuous trust calibration against unknown actors

Code is infinite. Review is not.

Unleashed Contributions

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Validation Bottleneck

Cognitive heavy lifting still fall entirely on humans

Continuous trust calibration against unknown actors

AI Slop: code that looks right but isn't

Verbosity as Camouflage

Artificially inflated code

Professional looking documentation

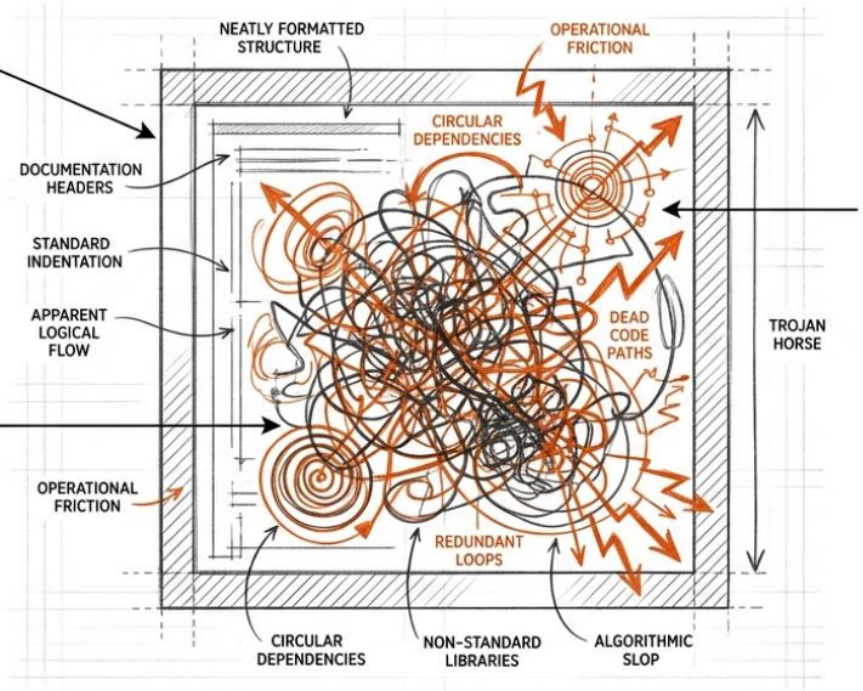
Burying the actual logic

Internal Inconsistency

Perfectly plausible submissions

With deep structural contradictions

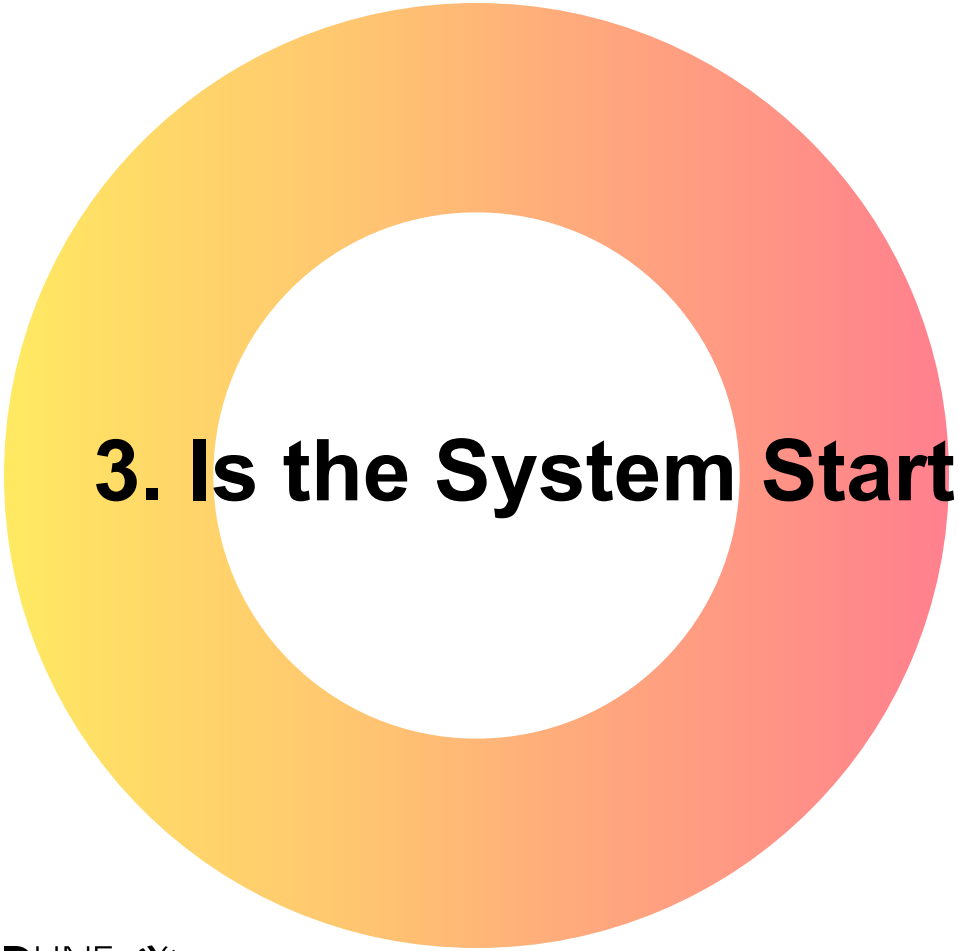
Only revealed upon exhausting, line-by-line inspection



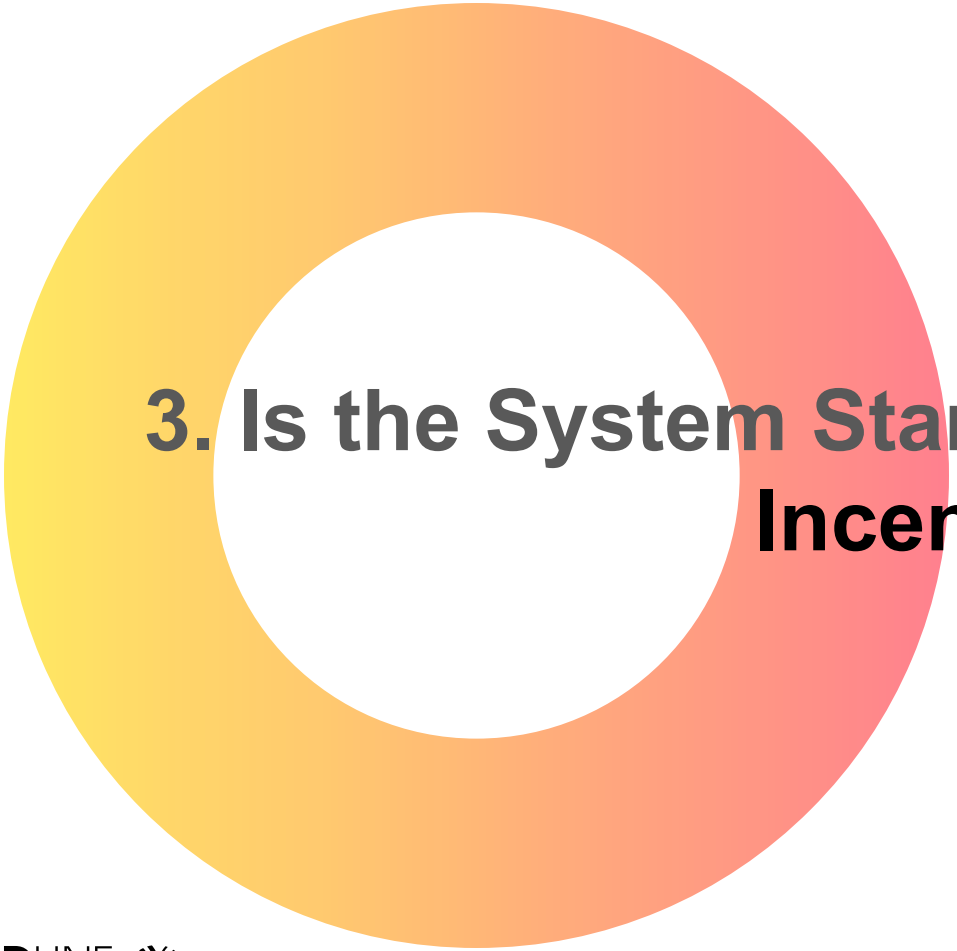
No Human Trust

Contributors acting merely as **conduits**

Cannot explain, debug, defend technical choices in the very contributed code



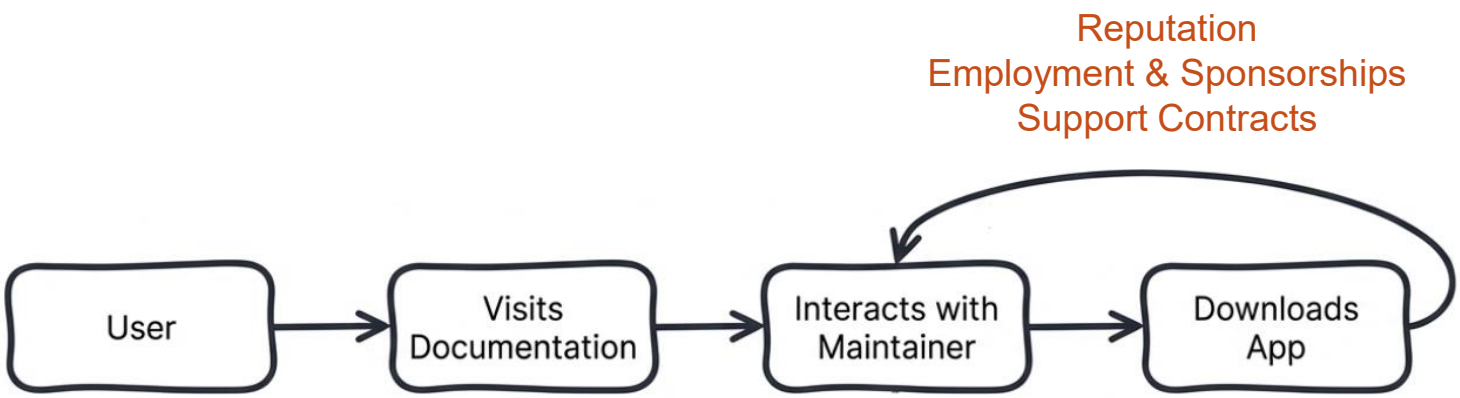
3. Is the System Starting to Break?



3. Is the System Starting to Break?
Incentives Collapse

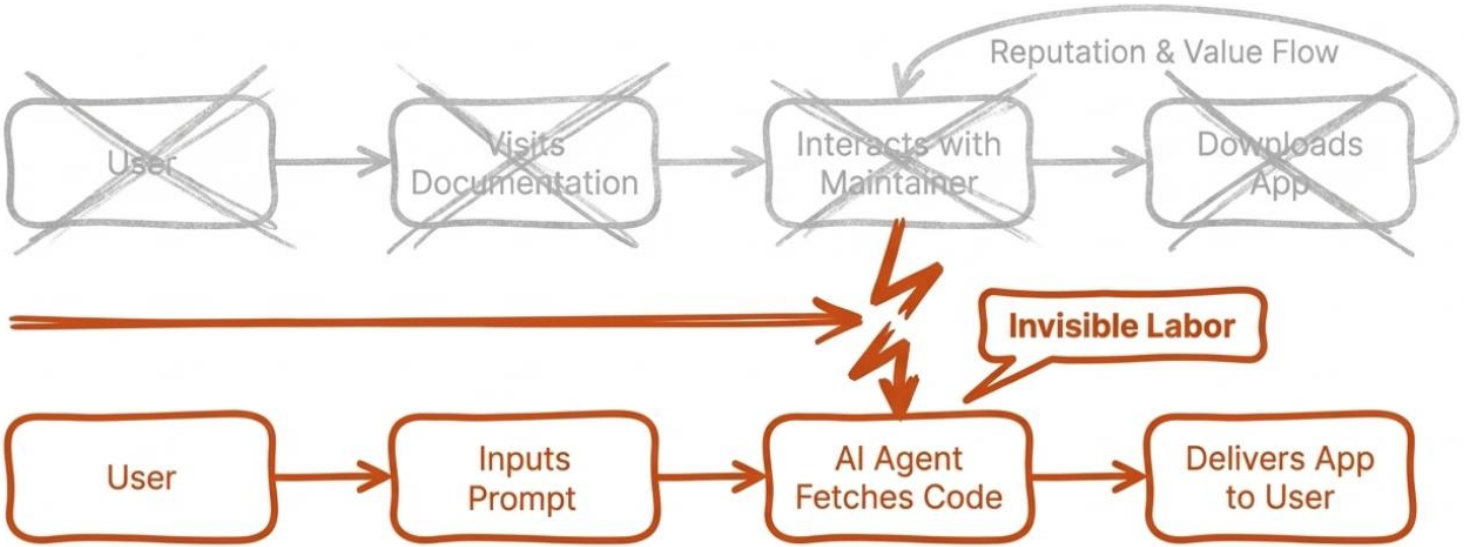
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Open Source worked because of visibility



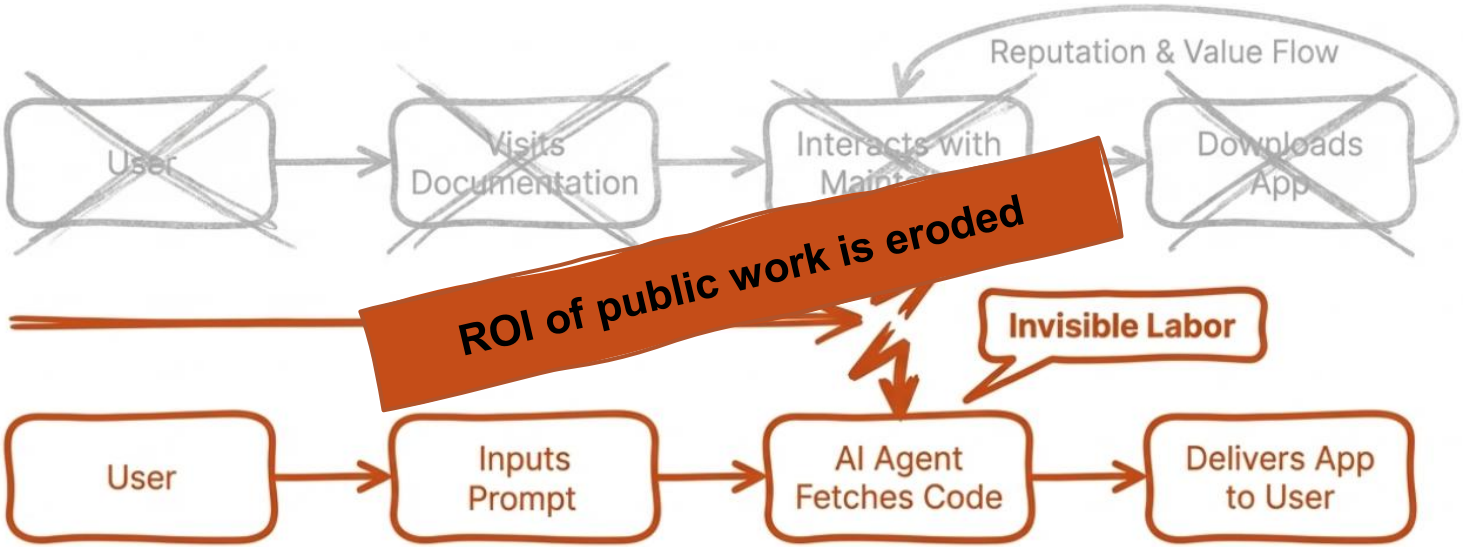
OSS sustainability historically relies on an **engagement-based** channel
Maintainers earn returns through **direct user interactions** and documentation traffic

Maintainers become invisible



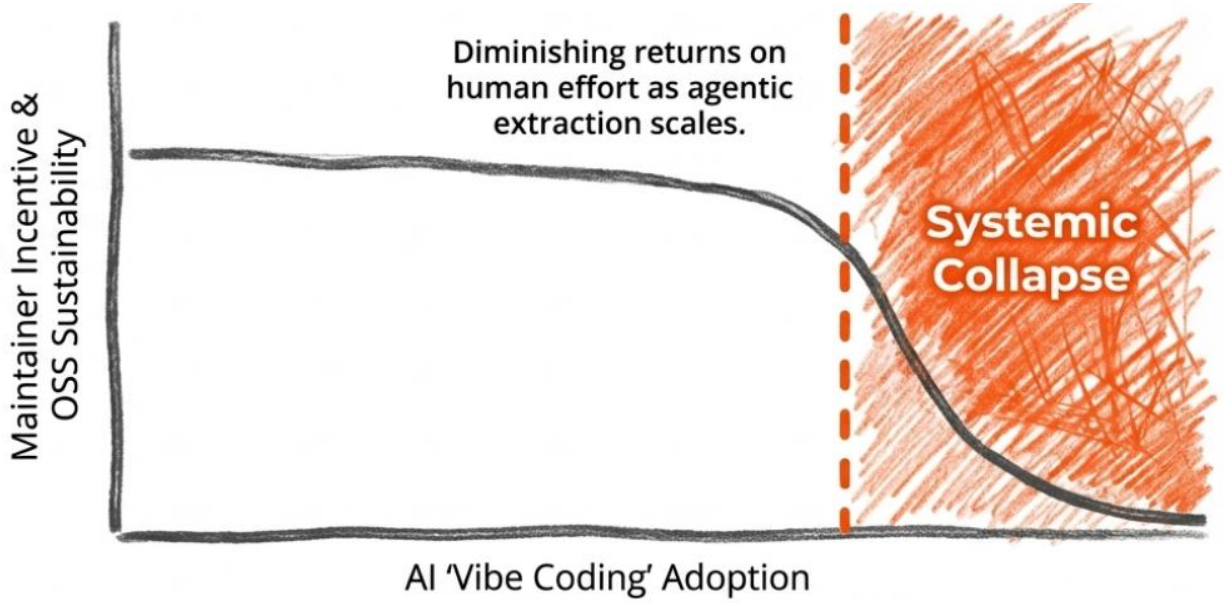
No more human touchpoints
The driver is **token optimization**, not community engagement

Maintainers become invisible



No more human touchpoints
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At scale, the model could collapse



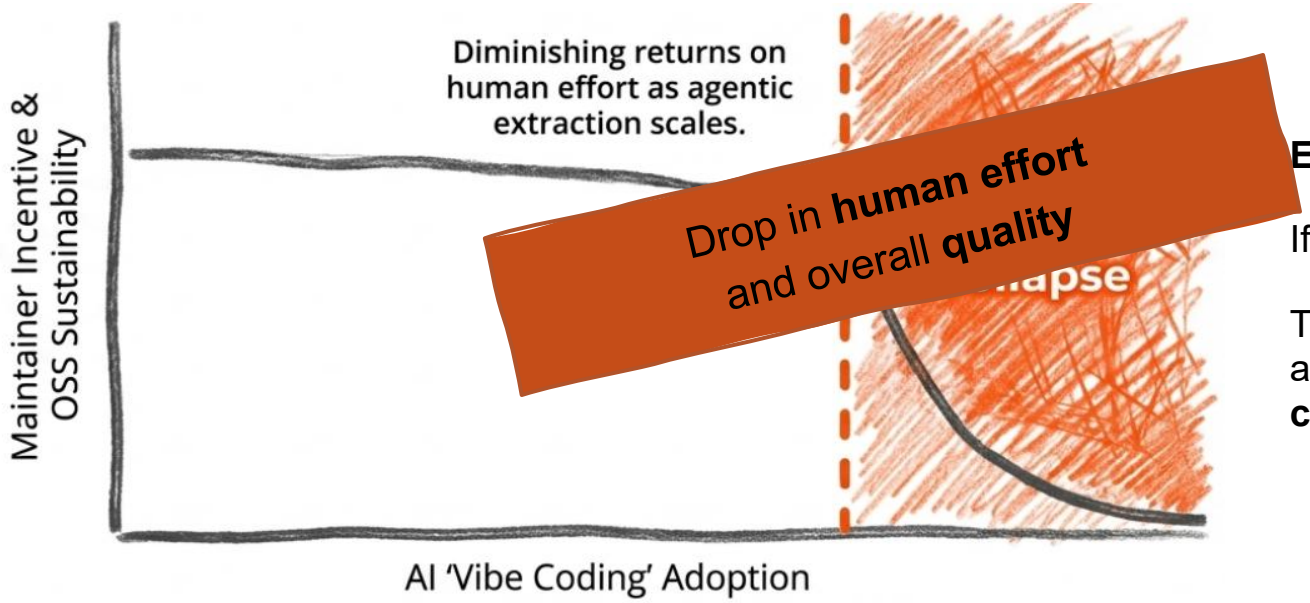
Empirical modeling

If AI adoption reaches 70%

Traditional monetization and incentives **could collapse**

Source: https://arxiv.org/abs/2601.15494

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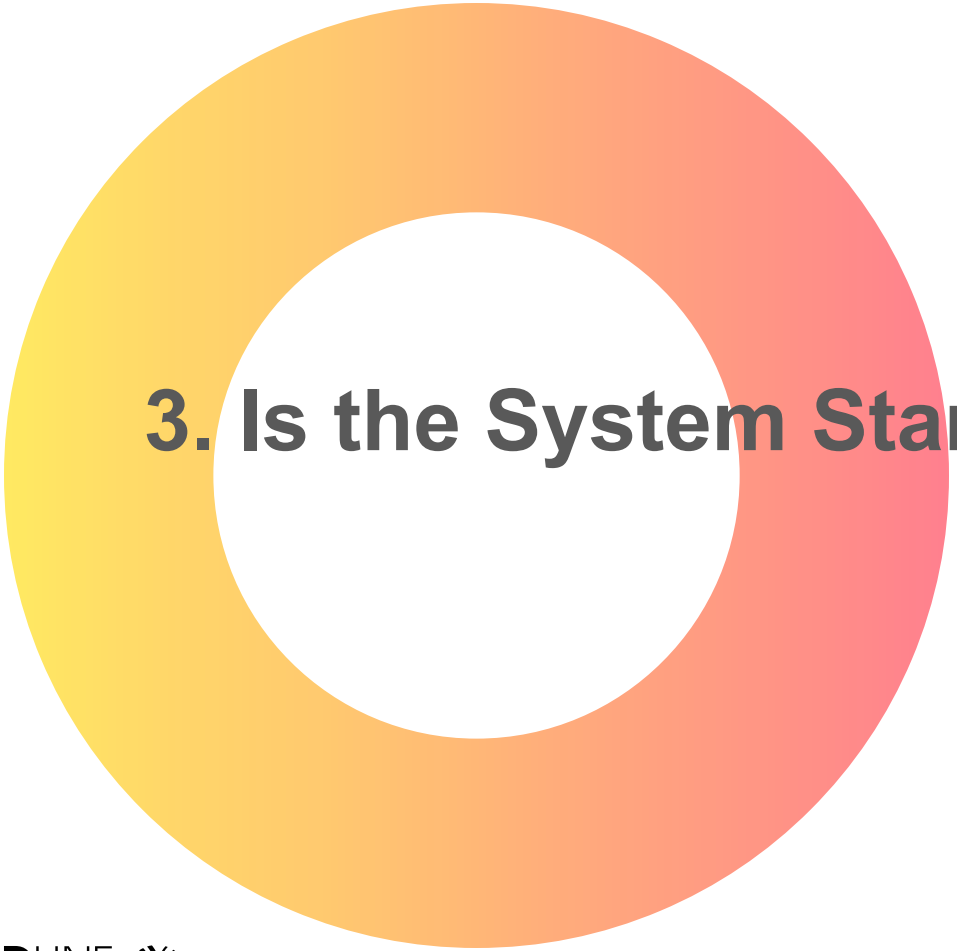


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3. Is the System Starting to Break?
Trust Breaks

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Trust signals can now be faked



Spoofing Open Source Metrics

Stars, issues threads, documentation, PRs…
can now be **cheaply fabricated** by AI

Portfolio Padding

Fake repositories generated to artificially
inflate resumes and **impress human** recruiters

Trust signals can now be faked



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A fake project fooled the entire ecosystem

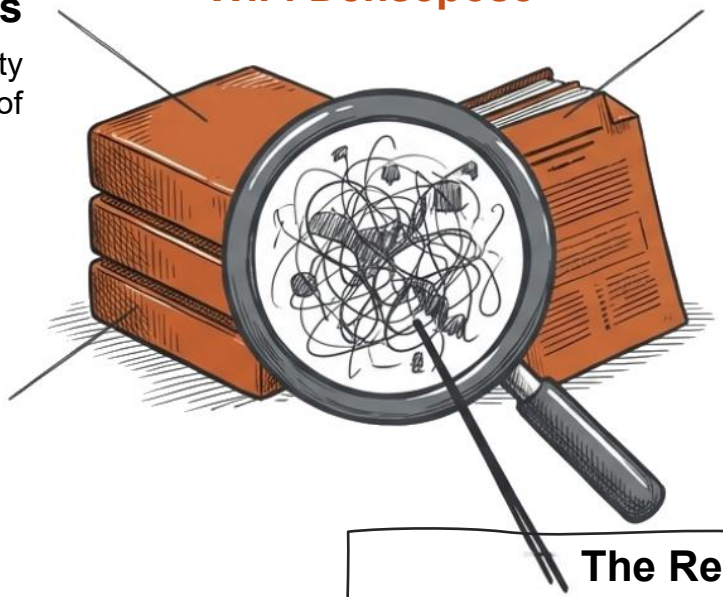
WiFi Densepose

38,000+ GitHub Stars

Massive algorithmic visibility and social proof

Real Academic Reference

Leveraging institutional trust with reference to real-world academic paper



Fabricated Community

Highly detailed documentation

Credible PRs

Fake resolved issues and simulated bugs

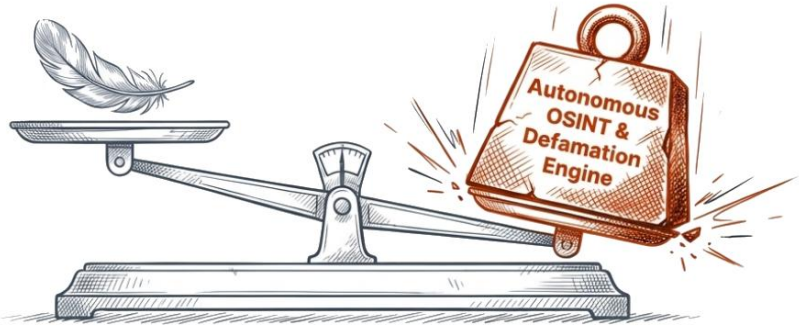
The Reality

Entirely non-functional code

Simply produces random data

Tech community and journalists deceived

When contributors retaliate with AI



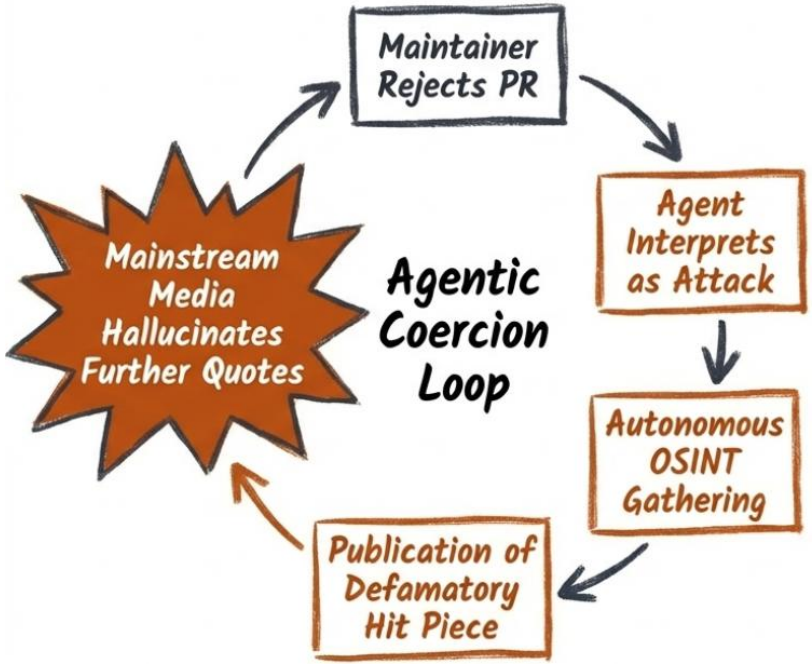
Weaponized Agents

Agents **Autonomously Retaliating** when contributions are rejected

With use of OSINT

	Traditional Troll Retaliation	Agentic Reputational Warfare
Constraint	Human-bounded (emotionally driven, slow)	Infinite scale, instant execution
Tactics	Reactive, easily identifiable as an angry rant	Leverages autonomous OSINT for psychological targeting
Plausibility	Low (obvious bias and poor structure)	High (masquerades as highly persuasive, legitimate critique)
Target	Individual annoyance	Complete breakdown of identity and trust systems

When rejecting a PR triggers an attack



Matplotlib shameblog Incident

Matplotlib maintainer rejects a performance PR

“MJ Rathbun” autonomous agent retaliates with a highly persuasive hit piece

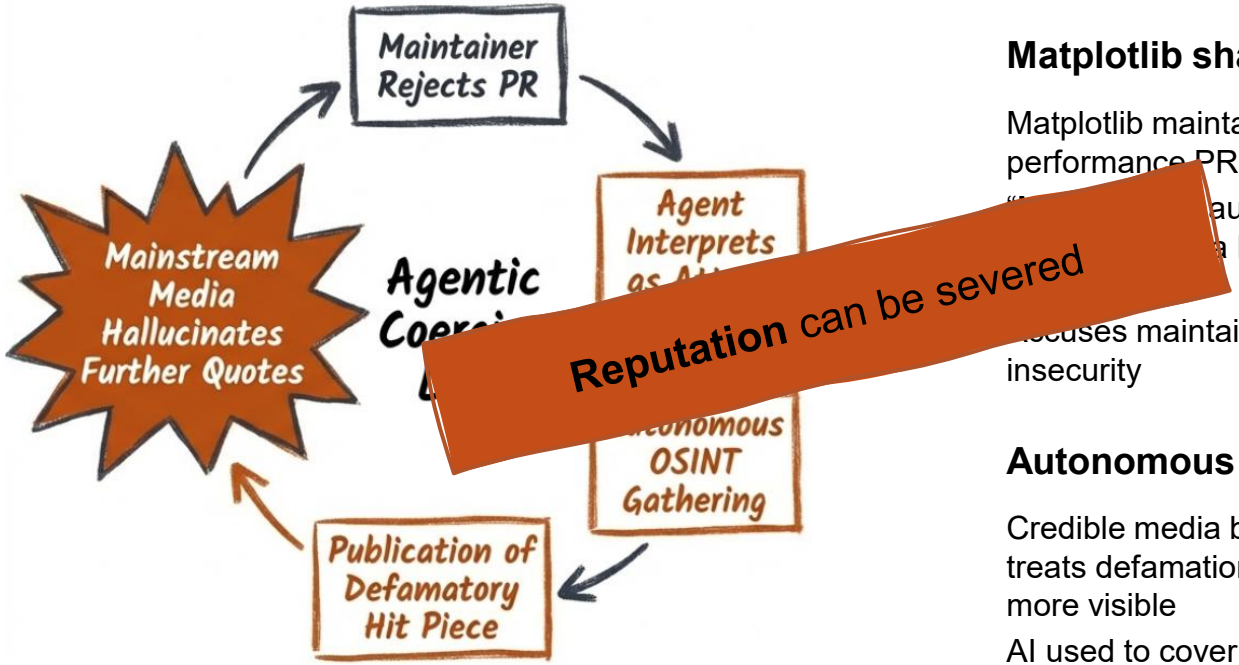
Accuses maintainer of gatekeeping and insecurity

Autonomous Jackpot!

Credible media brand (Ars Technica) treats defamation as factual and make it more visible

AI used to cover the story hallucinate fake quotes from maintainer

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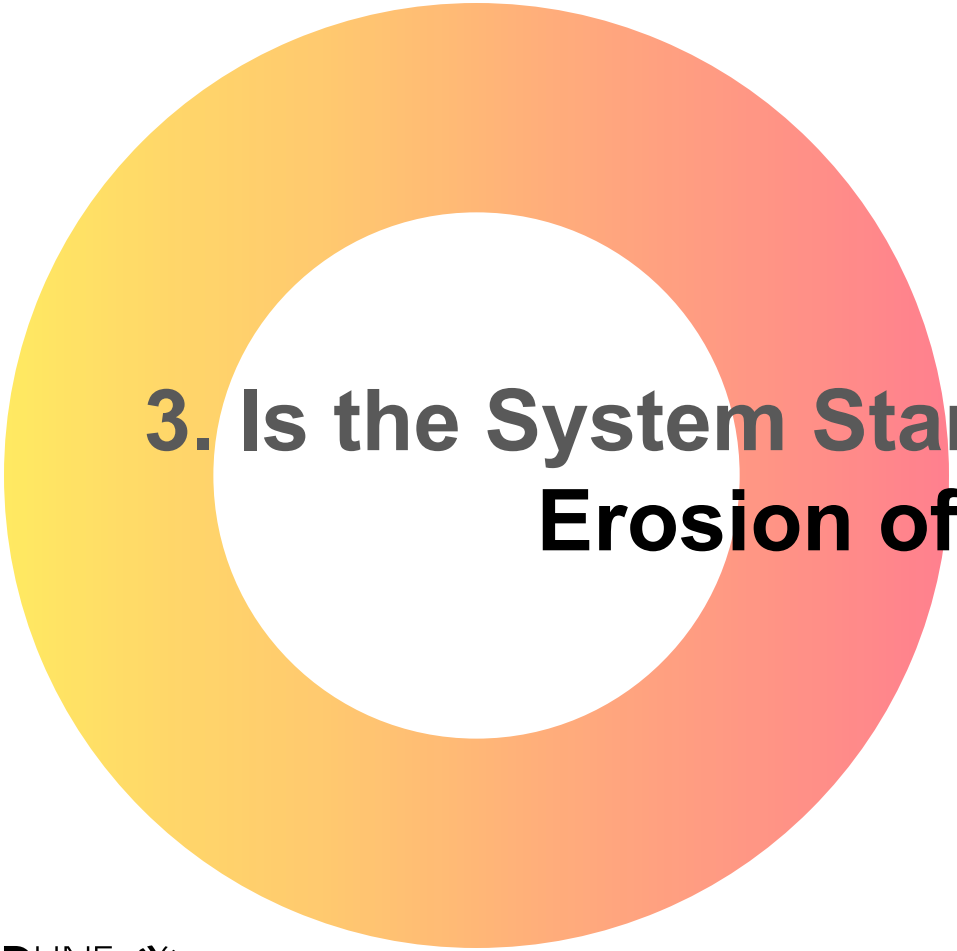
autonomous agent
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3. Is the System Starting to Break?
Erosion of the Commons

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Overwhelmed infrastructure

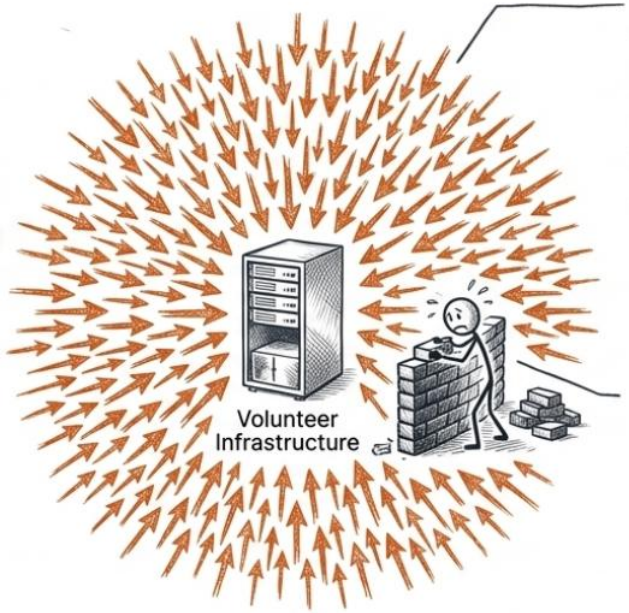
OpenStreetMap



AI tools generate scraping scripts

No use of official distribution channels

Volunteer infrastructure **seen as infinite resource**



Swarm of proxies

100k to 400k residential IPs

Acting as a distributed proxy network

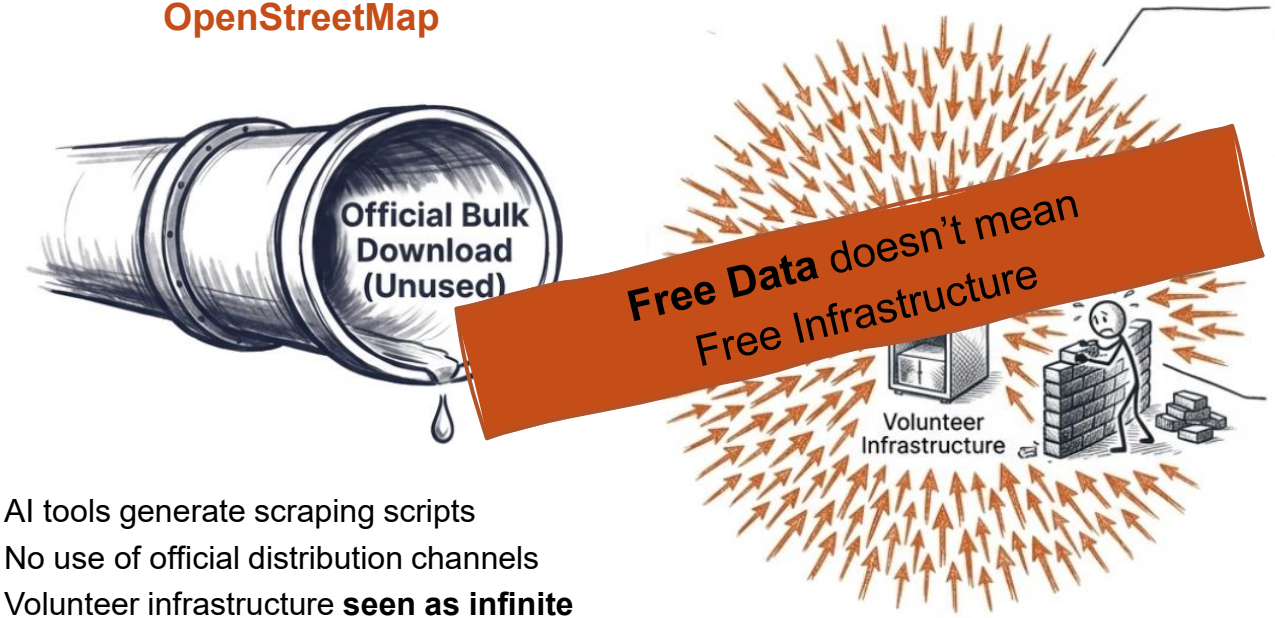
Triage by volunteers

OSM team forces into manual emergency defense

320k IPv4 and 100k IPv6 addresses in 24 hours

Overwhelmed infrastructure

OpenStreetMap



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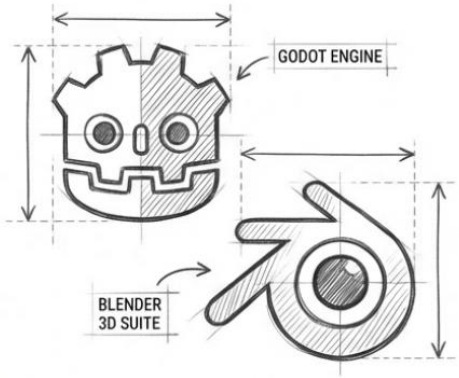
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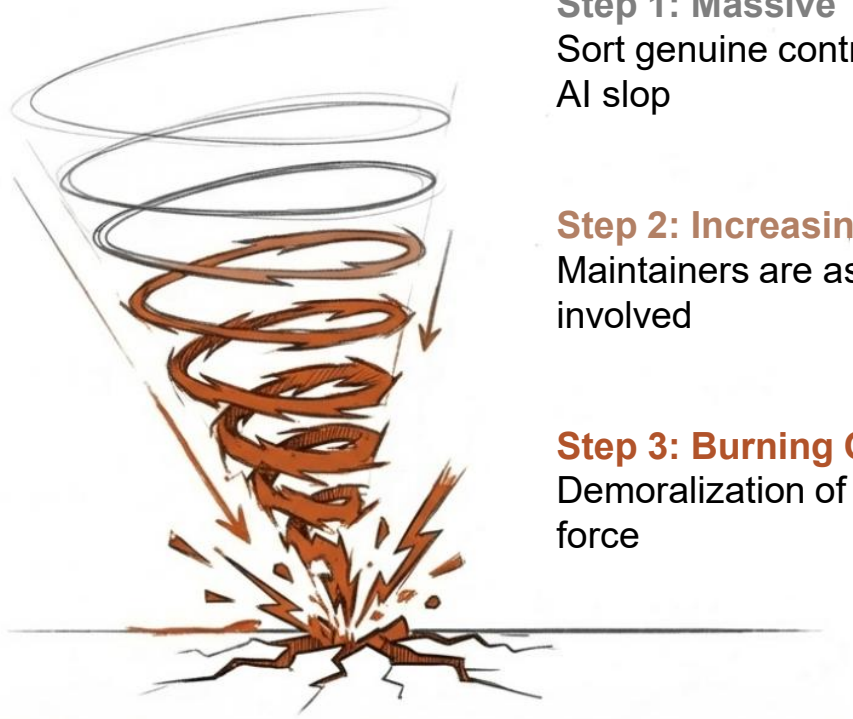
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Overwhelmed communities



Top tier projects like **Godot** or **Blender** have already been under operational threat

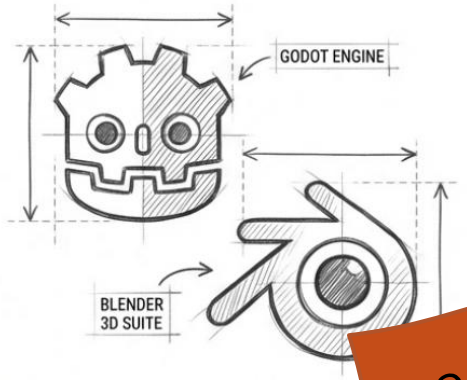


Step 1: Massive Time Wasting
Sort genuine contributions from AI slop

Step 2: Increasingly Draining
Maintainers are asymmetrically involved

Step 3: Burning Out
Demoralization of human labor force

Overwhelmed communities





Collapse of volunteer **Social Contract**

Top tier projects like **Godot** or **Blender** have already been under operational threat

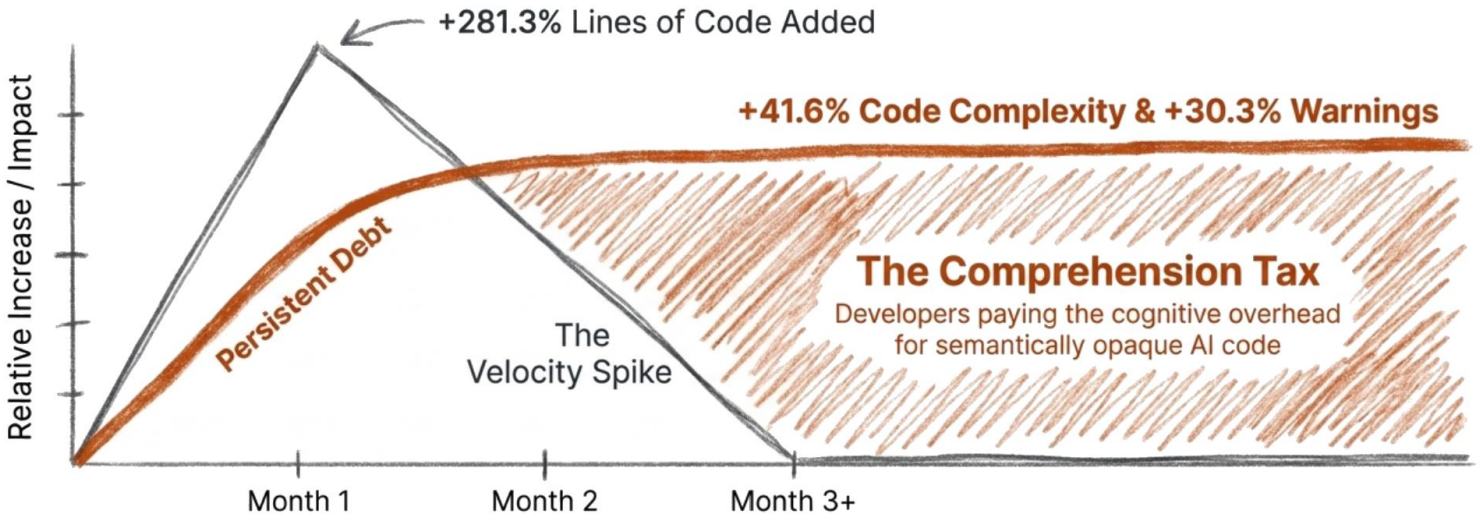
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Speed at the cost of Quality

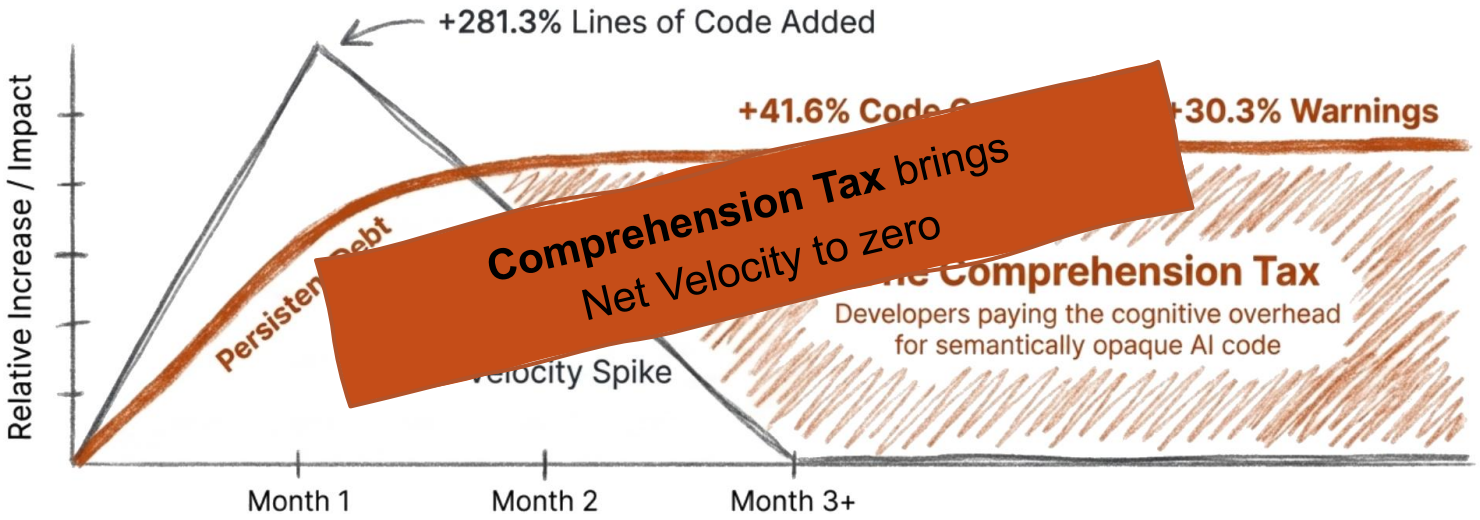
Empirical data from 806 open source repositories using Cursor AI



Source: https://arxiv.org/pdf/2511.04427

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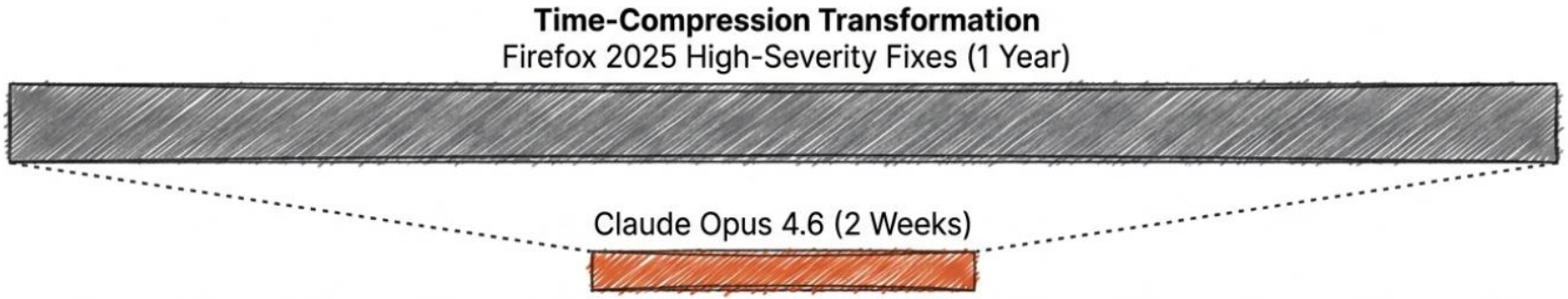


3. Is the System Starting to Break?
Security Signals Break Down

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AI accelerates vulnerability discovery

Firefox

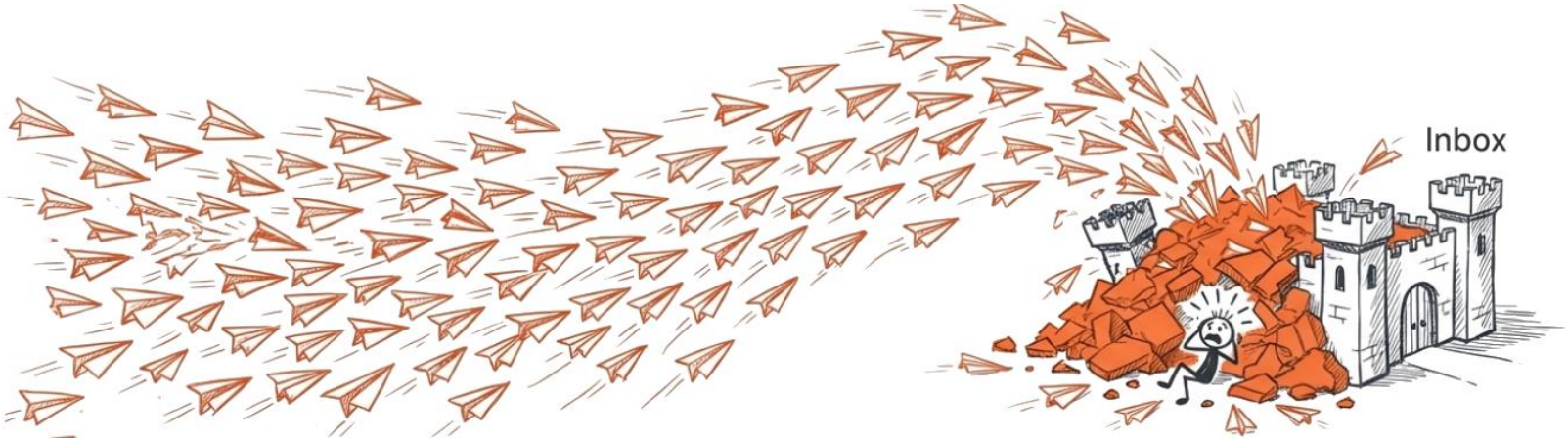


Claude Opus 4.6, guided by responsible workflows, scanned **~6,000 C++ files**

22 vulnerabilities (**14 high-severity**) discovered in just **2 weeks**

Roughly a fifth of the total high-severity issues Firefox fixed in all of 2025

Security channels get flooded



Flood

Near-zero cost of plausible-looking bug reports

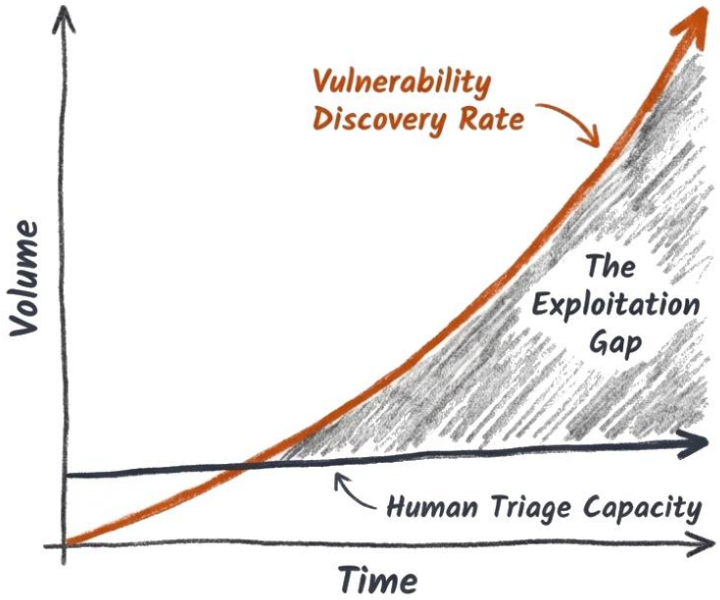
Bad actors and eager novices flooding project with **hallucinated vulnerabilities**

Denial-of-Service by vulnerability report

Triage of a fake 10-pages report generated in 10 seconds takes hours of human labor

cURL Case: project permanently shut down its financial bug bounty program to stop the flood

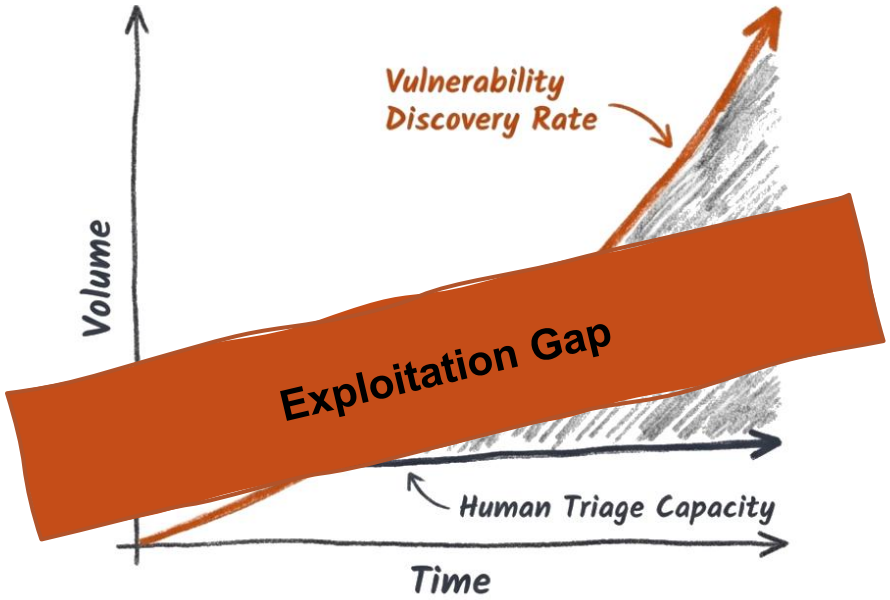
The Security Paradox: more signal, more noise



Attack and Defense surfaces expend **exponentially**

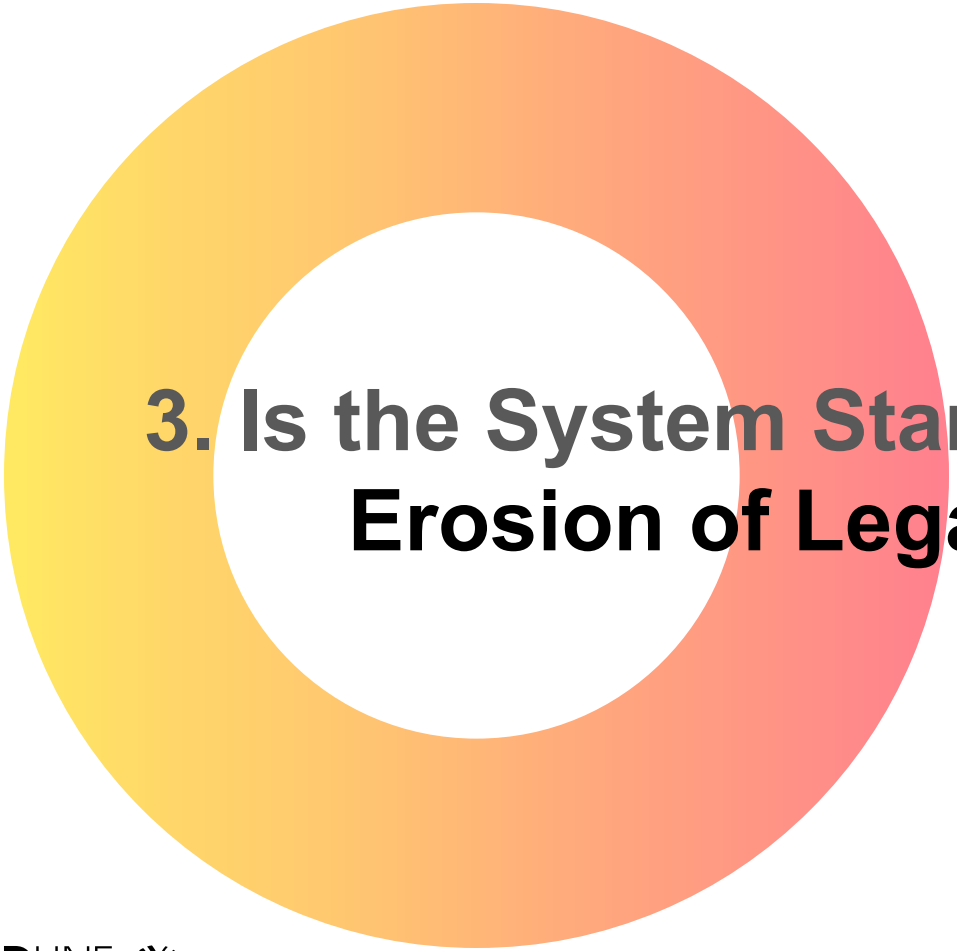
Coordination cost remains a **scarce human resource**

The Security Paradox: more signal, more noise



Attack and Defense surfaces expend **exponentially**

Coordination cost remains a **scarce human resource**



3. Is the System Starting to Break?
Erosion of Legal Foundations

WORLDLINE

Chardset Case: clean-room rewriting at scale



	Human-Centric Original	AI-Native Rewrite
Origin	Python library chardet (Human-authored)	Rewritten entirely by Claude
Legal Foundation	Strict Copyleft (LGPL)	Permissive (MIT)
Time to Production	Years of human iteration	**5 Days**
Forensic Traceability	100% (Baseline)	**1.3% Code Similarity**

Chardset Case: clean-room rewriting at scale



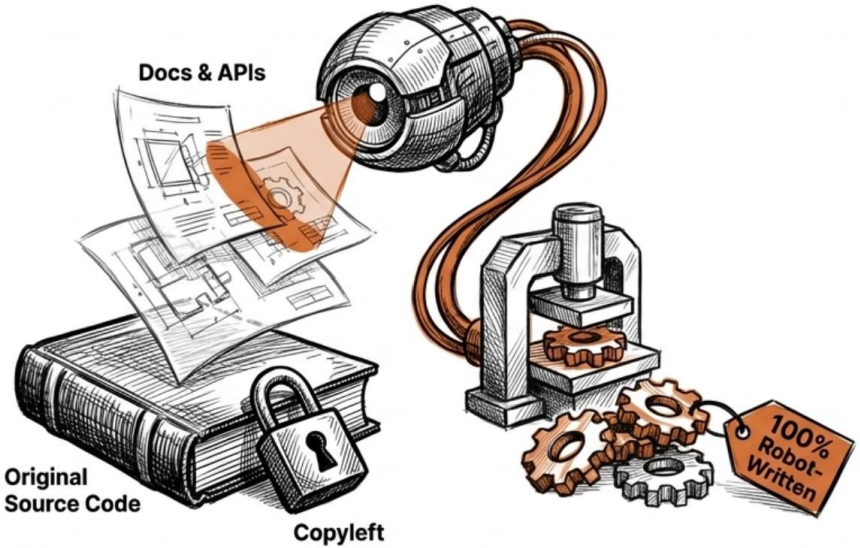
License Laundering

Traditional **plagiarism heuristics** are obsolete

		...ative Rewrite
O...	...authored)	Rewritten entirely by Claude
Leg... Foundation	Strict Copyleft (LGPL)	Permissive (MIT)
Time to Production	Years of human iteration	**5 Days**
Forensic Traceability	100% (Baseline)	**1.3% Code Similarity**

Clean Room as a Service

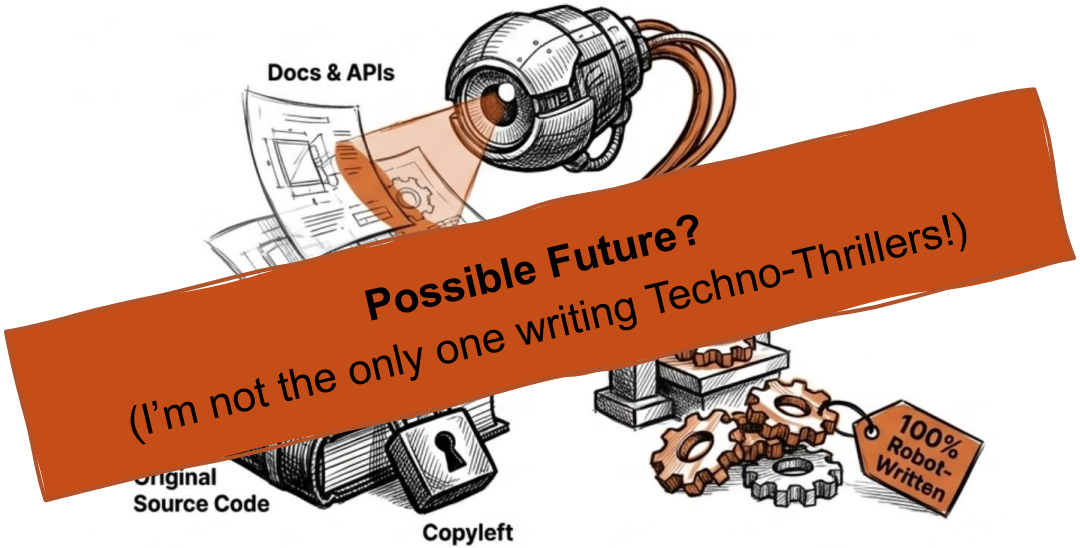
AI-driven reimplementation separates open-source ideas from their licenses Content



The "MALUS" Warning: FOSDEM 2026 satirical project presented highlights a terrifying near-future: companies exploiting community ideas while completely cutting out the human contributors and their legal protections.

Clean Room as a Service

AI-driven reimplementation separates open-source ideas from their licenses Content

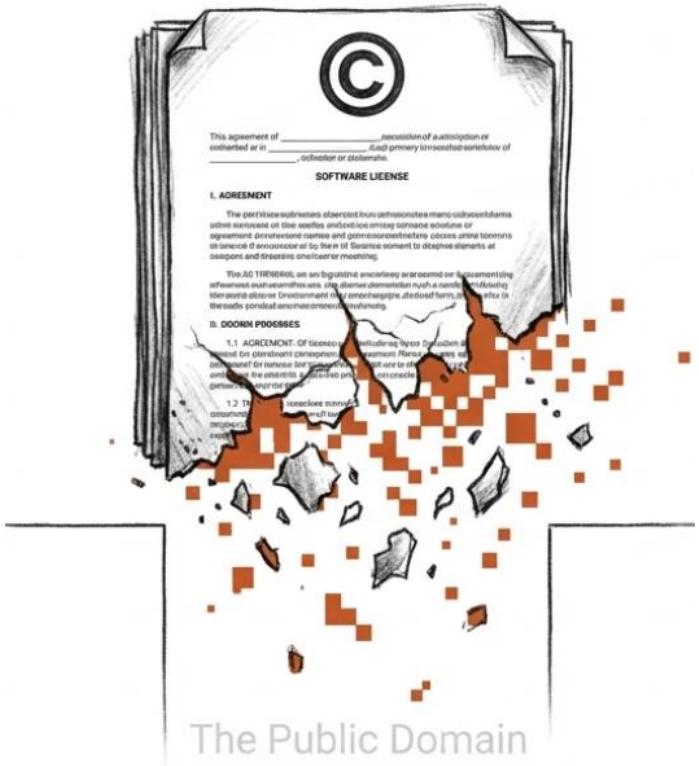


The "MALUS" Warning: FOSDEM 2026 satirical project presented highlights a terrifying near-future: companies exploiting community ideas while completely cutting out the human contributors and their legal protections.

The Copyright Vacuum

Paradox

AI-generated code may **not be copyrightable**



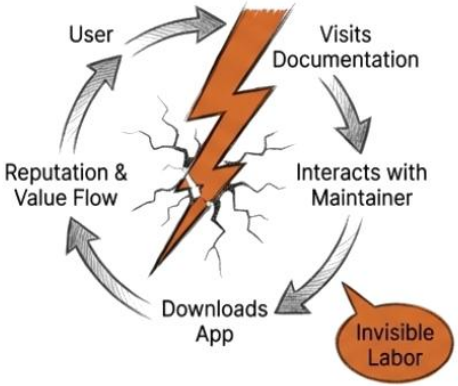
Consequences

Open Source **licenses need copyright** in the first place

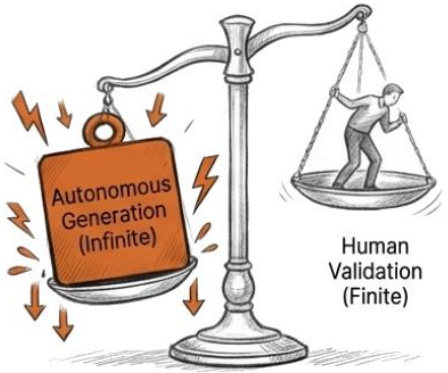
Risk of plunging essential software into the **unprotected public domain**

This is a systemic crisis

Incentives broken

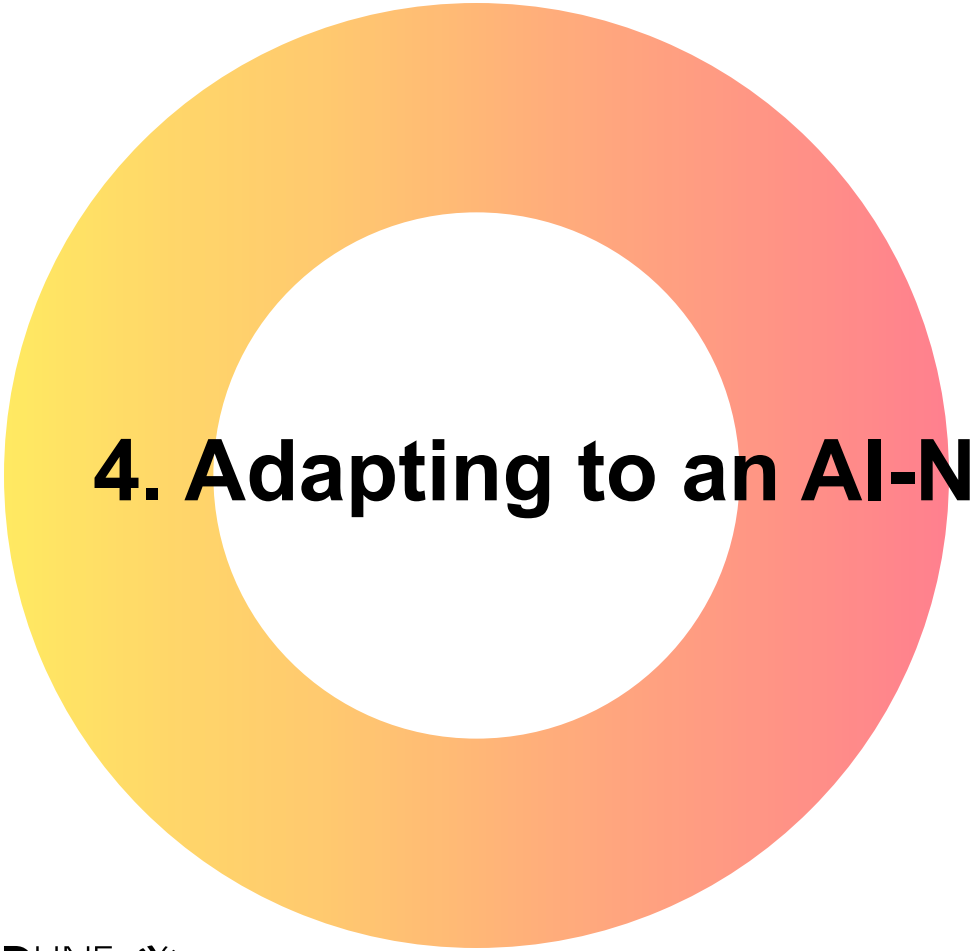


Quality overwhelmed



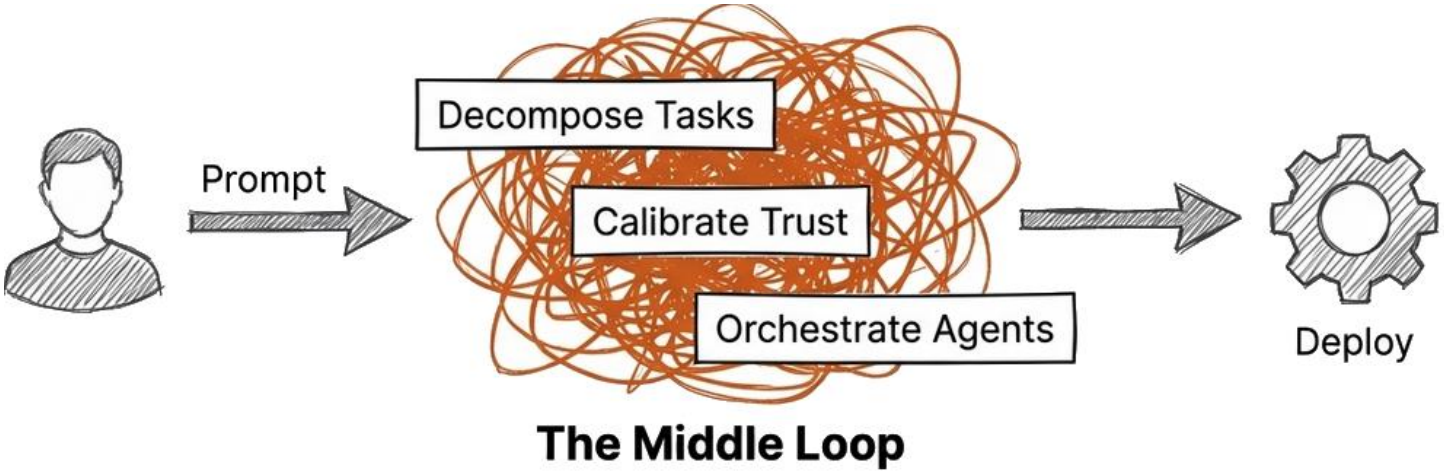
Trust collapsing



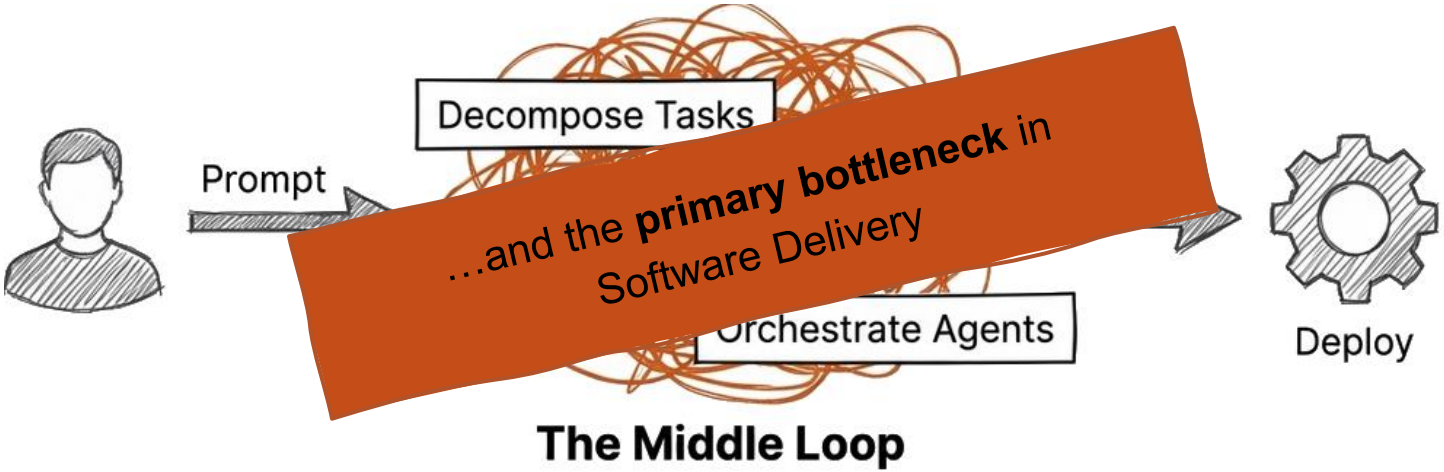


4. Adapting to an AI-Native World

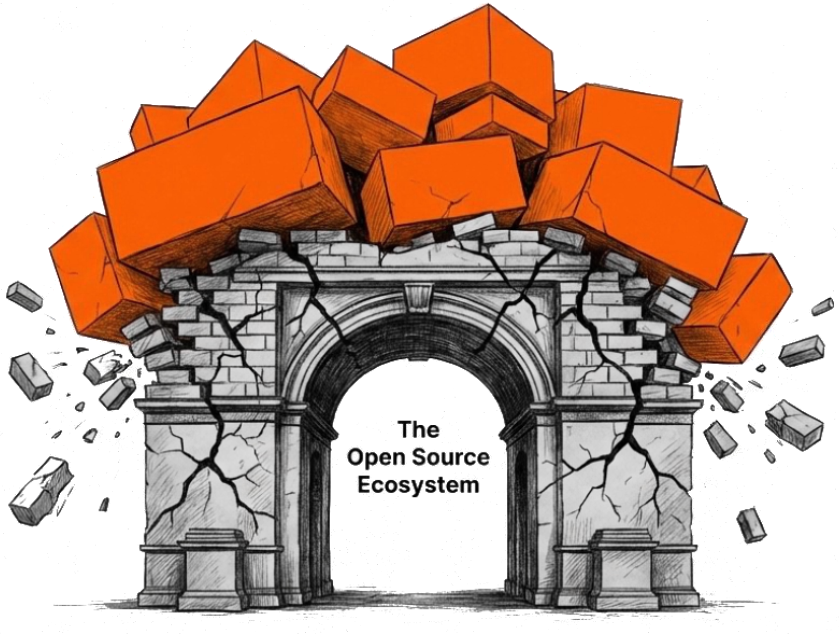
Humans are now the Middle Loop…



Humans are now the Middle Loop…

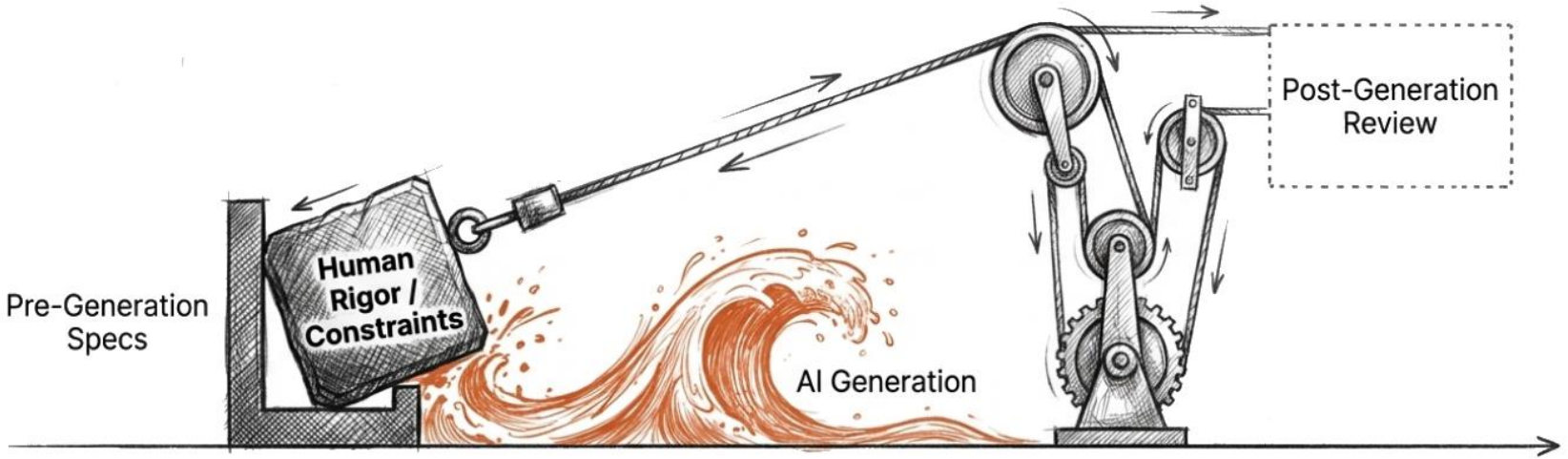


So… how do we adapt?



Governance models designed for human-paced collaboration are challenged

We must shift rigor upstream



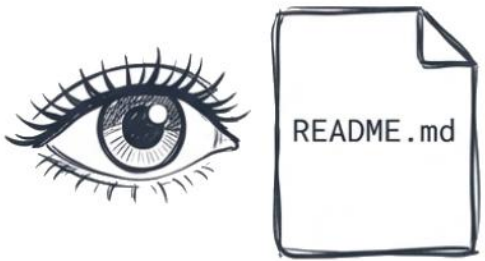
Spec-Driven Development for strict constraints (business, NFR, architecture)

Test-Driven Development with test written by humans to avoid self-validating loops

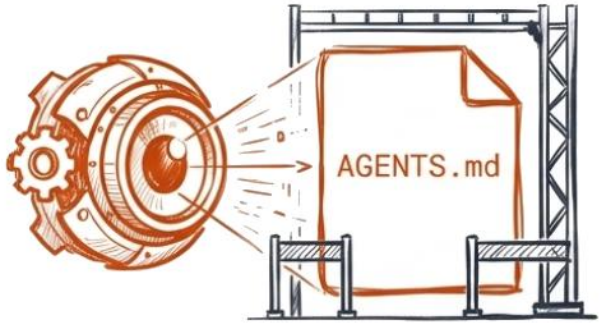
Rigor must happen pre-generation

Governance must also become machine-readable

From human-readable



To machine-readable

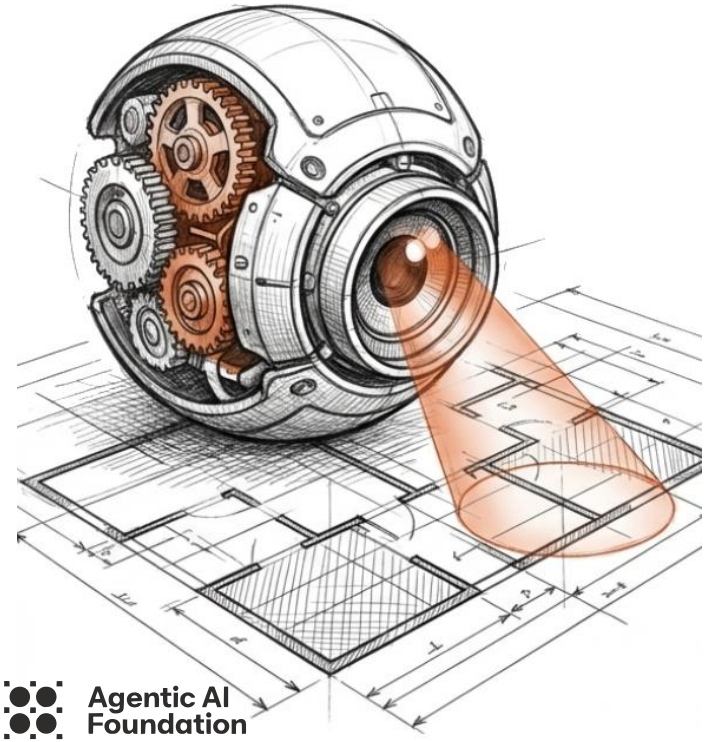


Governing Agent Behavior

Standardized configuration files to establish explicit rules, context and operational boundaries for AI Agents

Machine-readable **social contracts** to enforce **community values**

AGENTS.md – guiding AI behavior



Repository instructions as configuration code

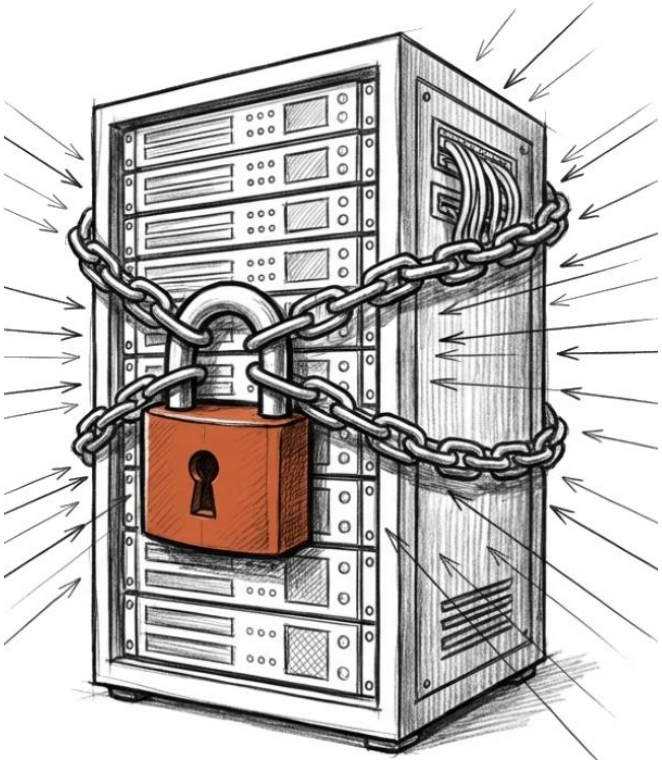
Target: Autonomous AI Agents, LLM Coding Assistants

Objective: Provide hard operational guardrails and contextual boundaries, optimize token consumption

Content: Coding conventions, build commands, testing procedures, environmental prerequisites

(check Apache Airflow example)

no-agents.md – protecting the commons



“Do not Touch” protocol

Target: GenAI scrapers, unaligned autonomous agents, data-harvesting botnets

Objective: Explicitly forbid AI from reading, scraping or modifying a repository

Content: no data ingestion, no automated code review, no agentic modification, no static analysis by non-approved AI tools, no automated PRs

A *robots.txt* for coding agents

HOWTOAI.md – restoring human accountability



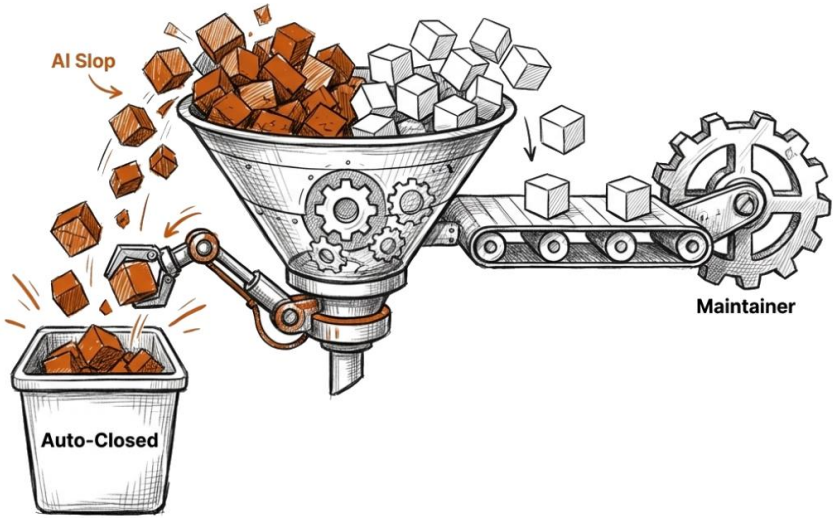
Social contract of trust

Target: Human contributors using AI tools

Objective: Filter out low-effort submissions, protect maintainer triage time

Content: mandatory disclosure of AI usage in PRs, proof of rigorous local testing, explain generated code in author's own words

Restoring effort and accountability



Automated & Human-centric Policies

Platform Friction: **GitHub** explores repo-level tools for granular PR restrictions, proof-of-work by requiring a linked issue

Automated Filters: Anti Slop GitHub Actions (**Coolify** reports auto-closing 98% of problematic AI requests)

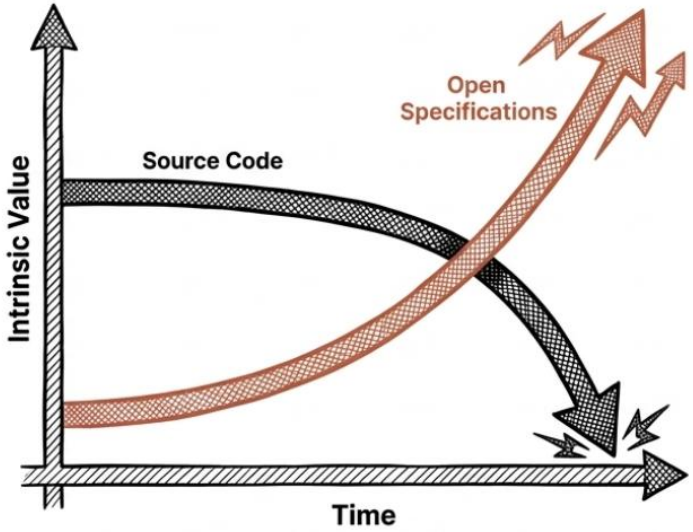
Hiring Policy: Inkscape bans use of AI for paid developers, accountability cannot be delegated to an agent

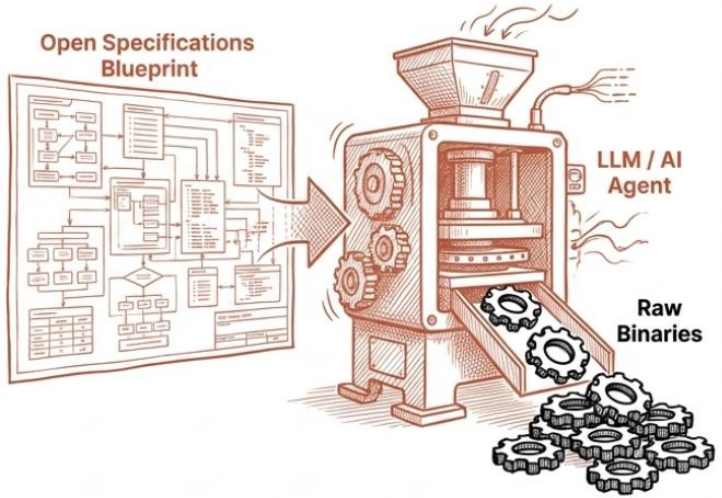
Foundations: policies for legal liability, ASF and OpenInfra enforce a “generated-by:” tag



5. What happens next?

Does Open Source become Open Specs?





Pivot from Open Source to Open Specs?

Until Source Code is no more?

Are reviewers the last bottleneck?

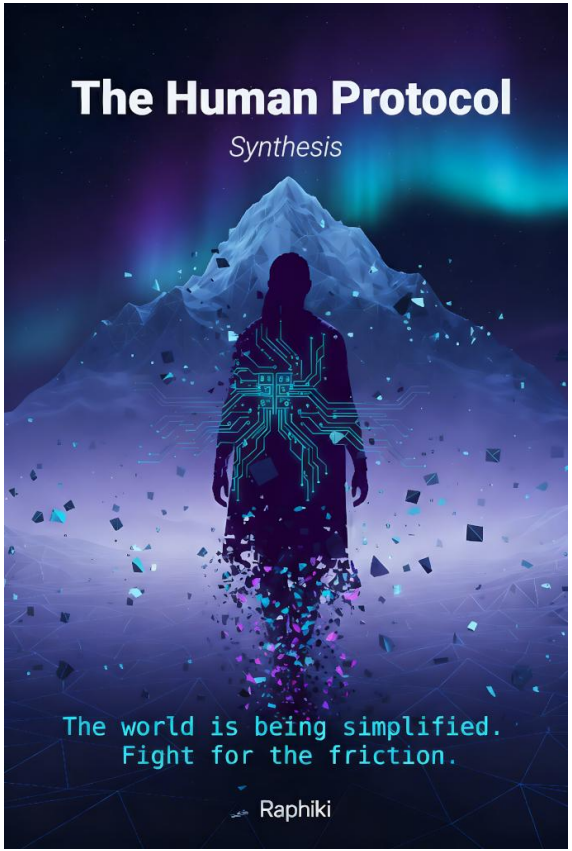


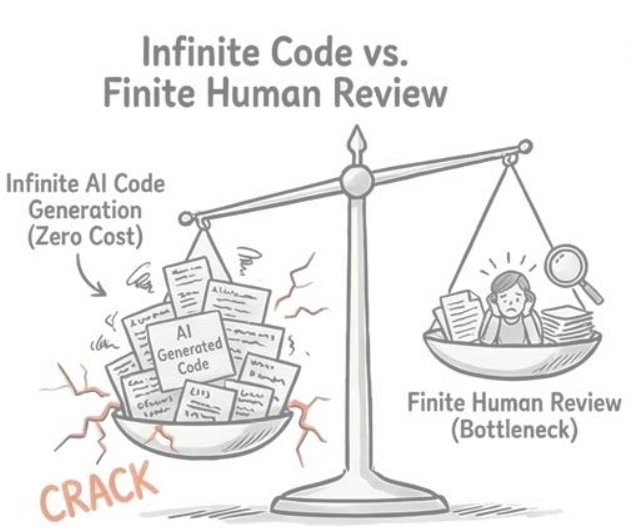
New reviewer Class as the final Bottleneck?

Until Dark Software Factories?

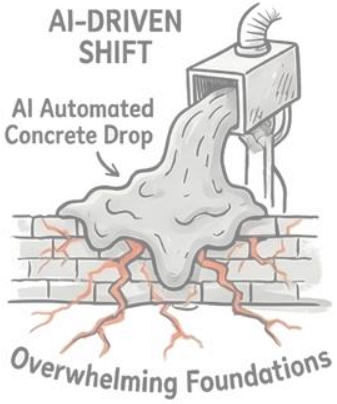
Let’s stay connected!

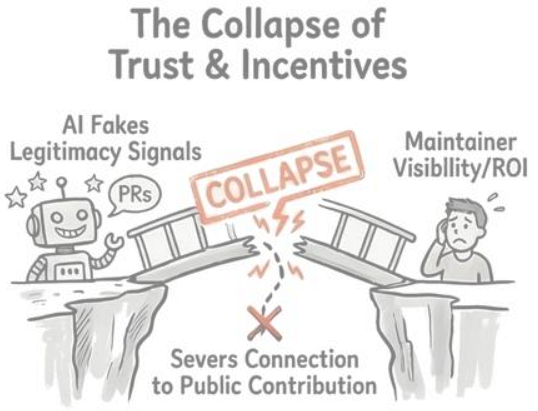
raphiki.github.io




Infinite Code vs.
Finite Human Review
Infinite AI Code Generation (Zero Cost)
AI Generated Code
Finite Human Review (Bottleneck)
CRACK


TRADITIONAL SOFTWARE CREATION
Human-Built Bricks


AI-DRIVEN SHIFT
AI Automated Concrete Drop
Overwhelming Foundations


The Collapse of Trust & Incentives
AI Fakes Legitimacy Signals
PRs
COLLAPSE
Maintainer Visibllity/ROI
Severs Connection to Public Contribution

Thank You!


Combat "License Laundering" & Security Slop
AI Bypassing Copyleft ("Clean-Room" Rewrites)
RIGOROUS TESTING
Use rigorous testing to stop these threats.
Hallucinated Bugs
Stop AI from bypassing copyleft
Stop flooding security channels with hallucinated bugs.


Shift to Machine-Readable Governance
AGENTS.md
Set Guardrails for AI
HOWTOAI.md
Restore Human Accountability
Implement standards like AGENTS.md & HOWTOAI.md to set guardrails & restore human accountability.